Optimal spectral supersaturation threshold for cliques

Hongzhang Chen, Yongtao Li

arxiv:2610.06038v1; CC BY4.0

Editorial note. Complete original mathematical paper93–2303 with all supporting proofs, native inline figures, author sharpness constructions and embedded bibliography retained. Family-specific main arguments appear in their original section prose as well as proof environments; no new proof environment or mathematical explanation is invented.

Editorial note. Only the selected uniform family theorem is counted. Shared triangle specializations, second-order spectral estimate, continuous consequence, sharpness construction and equality refinements are not extra counts. The other principal family remains complete context.

Editorial note. Original acknowledgement2059–2062 says language-model tools were used at the early exploratory stage to brainstorm candidate proof strategies, and suggestions served as informal inspiration; authors say they developed arguments and wrote/verified all proofs. Exact disclosure retained as author statement, without agent certification, AI-free labeling or inference of selected-core AI discovery.

Editorial note. Authors retain responsibility for mathematics. No proof repair, added hypotheses, independent proof review, counterexample search or AI dependency-chain investigation.

Editorial note. Actual r001 ctex dependency was triggered solely by three non-rendering Chinese author comments at21/1261/1375. In source0 only these comment lines are replaced by ASCII comments, retaining all2304 logical lines and all rendered mathematics/figures. Original author TeX/archive and failed r001 preserved. No Chinese font/package installation or parser change.

Allowed standard prerequisites (editorial).

- Spectral Turan stability, color-critical supersaturation and Perron eigenvector estimates. Original fixed-family/r/k, sufficiently-large n, epsilon and q-range hypotheses, exact c(n,F), Y and extremal-family definitions retained. Original general Fang-Li-Lin-Ma structural/small-unspecified-constant theorem is cited background.. Complete native local spectral perturbation, graph-counting, extremal construction and family-specific proof supplied; no independent construction or reconstruction of external cited proofs.

1 Introduction

The Turán-type problems ask for the maximum number of edges in a graph with no copy of a given graph F. Let $T_{n,2}$ be the complete balanced bipartite graph. The well-known Mantel theorem says that if G is an n-vertex graph with no triangle, then $e(G) \leqslant e(T_{n,2})$, with equality if and only if $G = T_{n,2}$; see [2]. A central theme in extremal graph theory is the supersaturation problem: once a graph has more edges than the Turán number of F, it must contain not just one copy of F but many, and one asks for the precise count. The oldest instance is the Erdős–Rademacher problem on triangles, whose study over several decades, from Erdős [8, 9], Lovász–Simonovits [24, 25] to the complete solution of the clique density problem by Razborov [37], Liu–Pikhurko–Staden [20], Nikiforov [32] and Reiher [38], has produced some of the most influential methods in the field.

Erdős and Rademacher (see [8, 9]) extended Mantel's theorem by showing that if $e(G) > e(T_{n,2})$, then G contains at least $\lfloor n/2 \rfloor$ triangles. Moreover, Erdős [8] showed that there exists a constant $\delta > 0$ such that if n is sufficiently large and $1 \leqslant q < \delta n$, then every n-vertex graph with $\lfloor n^2/4 \rfloor + q$ edges contains at least $q\lfloor n/2 \rfloor$ triangles. Furthermore, Lovász and Simonovits [24] proved that for $1 \leqslant q < n/2$, if G is an n-vertex graph with $\lfloor n^2/4 \rfloor + q$ edges, then G contains at least $q\lfloor n/2 \rfloor$ triangles. The range $q < n/2$ is best possible. Indeed, let n be even and $q = n/2$. Then add $q+1$ extra edges to the larger partite set of $K_{\frac{n}{2}+1,\frac{n}{2}-1}$ in such a way that no triangle is formed among these $q+1$ edges. This yields a graph with $n^2/4+q$ edges and $(q+1)(n/2-1) = n^2/4-1$ triangles, which is fewer than $q\lfloor n/2 \rfloor$. We refer the reader to [42, 23, 1, 26] for recent developments.

A graph is called *color-critical* if it has an edge whose removal lowers its chromatic number. This large family, which includes cliques and odd cycles, plays a central role in extremal graph theory. Let $T_{n,r}$ denote the n-vertex complete balanced r-partite graph. Simonovits [39] proved that for every color-critical graph F with chromatic number $\chi(F) = r+1$, if n is large enough and G is an n-vertex graph with no copy of F, then $e(G) \leqslant e(T_{n,r})$, with equality if and only if $G = T_{n,r}$. Extending this result, Mubayi [27] proved a supersaturation theorem for all color-critical graphs. Let $c(n,F)$ denote the smallest number of copies of F created by adding an edge to $T_{n,r}$. Explicitly, $c(n,K_{r+1}) = (1+o(1))(n/r)^{r-1}$ and $c(n,C_{2k+1}) = (1+o(1))(n/2)^{2k-1}$. In particular, both reduce to $\lfloor n/2 \rfloor$ when $r=2$ and $k=1$, recovering the Erdős–Rademacher count for triangles.

Theorem 1.1 (Mubayi [27]). *Let F be a color-critical graph with $\chi(F) = r+1$. There exists a constant $\delta_F > 0$ such that for all sufficiently large n, and for every integer q with $1 \leqslant q \leqslant \delta_F n$, if G is an n-vertex graph with $e(G) \geqslant e(T_{n,r}) + q$, then G contains at least $q\,c(n,F)$ copies of F.*

Let $\mathcal{T}_{n,r,q}$ be the family of graphs obtained from the Turán graph $T_{n,r}$ by adding q edges. Pikhurko and Yilma [36] later sharpened Theorem 1.1 by describing the extremal graphs, showing further that under the hypothesis of Theorem 1.1, if G contains the fewest copies of F, then $G \in \mathcal{T}_{n,r,q}$. Throughout this paper, we write $\delta_{\mathrm{edge}}(F)$ for the *largest constant* δ_F for which Theorem 1.1 holds; this determines the optimal range of q in the edge setting. The problem of determining $\delta_{\mathrm{edge}}(F)$ for cliques F is a well-known question of Erdős, which was solved by Lovász and Simonovits [25].

Theorem 1.2 (Lovász–Simonovits [25]). *Let $r \geqslant 2$ be an integer. Then*

$$\delta_{\mathrm{edge}}(K_{r+1}) = \frac{1}{r}.$$

Apart from cliques, Pikhurko and Yilma [36] provided a powerful approach that allows one to determine the exact threshold $\delta_{\mathrm{edge}}(F)$ for a large number of color-critical graphs F, including odd cycles, cliques with one edge removed, and complete bipartite graphs plus an edge.

Theorem 1.3 (Pikhurko–Yilma [36]). *For every $k \geqslant 1$, we have $\delta_{\mathrm{edge}}(C_{2k+1}) = 1/2$; let $r \geqslant 2$ and $K_{r+2} - e$ be obtained from K_{r+2} by removing an edge. Then $\delta_{\mathrm{edge}}(K_{r+2} - e) = (r-1)/r^2$.*

1.1 Spectral supersaturation problems

Let $\lambda(G)$ be the spectral radius of the adjacency matrix of G, and let $\#F(G)$ be the number of copies of F in G. Spectral Turán theory replaces the edge count $e(G)$ by the spectral radius $\lambda(G)$ and looks for analogues of classical Turán-type results; see the survey [33]. In 1986, Wilf [41] showed that every K_{r+1}-free graph G on n vertices satisfies $\lambda(G) \leqslant (1-\frac{1}{r})n$; see, e.g., [29, 30, 11] for related results. The Simonovits theorem for color-critical graphs was generalized by Nikiforov [31]: if F is a color-critical graph with $\chi(F) = r+1$, and G is an n-vertex F-free graph with n large, then $\lambda(G) \leqslant \lambda(T_{n,r})$, with equality if and only if $G = T_{n,r}$. This spectral version implies the edge version in a non-trivial way; see [22]. Spectral supersaturation asks how many copies of F are forced in G once $\lambda(G)$ passes the spectral threshold $\lambda(T_{n,r})$. Counting substructures under spectral conditions

has attracted much attention in recent years. In 2007, Bollobás and Nikiforov [3] proved that

$$\#K_{r+1}(G) \geqslant \Big(\frac{\lambda(G)}{n} - 1 + \frac{1}{r}\Big)\frac{r(r-1)}{r+1}\Big(\frac{n}{r}\Big)^{r+1}. \tag{1}$$

This bound turns the spectral radius into a lower bound on the number of $(r+1)$-cliques, and it is exactly the inequality that our forthcoming Theorem 1.7 sharpens just above the Turán threshold. In 2023, Ning and Zhai [34] proved that if G is a graph on n vertices with $\lambda(G) \geqslant \lambda(T_{n,2})$, then G contains at least $\lfloor \frac{n}{2} \rfloor - 1$ triangles, unless $G = T_{n,2}$. In recent years, a number of results on spectral supersaturation have been obtained, including triangles [15, 45, 6], 4-cycles [35, 16], books [43, 44, 46, 40], color-critical graphs [12, 13, 14, 5] and some classical bipartite graphs [19].

To find a spectral analogue of Theorem 1.1, one needs a candidate graph such that $\lambda(G)$ above the candidate could force at least $q\,c(n,F)$ copies of F. The natural choice is $Y_{n,r,q}$, the graph obtained from $T_{n,r}$ by adding q *pairwise disjoint edges* inside one largest part. Among all graphs built from $T_{n,r}$ by adding q edges, $Y_{n,r,q}$ attains the smallest spectral radius; this was proved for all $r \geqslant 2$ and $q \leqslant \frac{n}{100r}$ in [12]. Thus $\lambda(G) \geqslant \lambda(Y_{n,r,q})$ is the *weakest* spectral condition under which one can hope to force $q\,c(n,F)$ copies. For triangles, Li, Feng and Peng [15] established the first result of this kind: if $1 \leqslant q \leqslant \frac{1}{9}\sqrt{n}$ and G is an n-vertex graph with $\lambda(G) \geqslant \lambda(Y_{n,2,q})$, then $\#K_3(G) \geqslant q\lfloor n/2 \rfloor$. For color-critical graphs, Fang, Li, Lin and Ma [12] proved a general result.

Theorem 1.4 (Fang–Li–Lin–Ma [12])**.** *Let F be a color-critical graph with $\chi(F) = r+1$. There exists $\delta_F > 0$ such that if n is sufficiently large, $1 \leqslant q \leqslant \delta_F\sqrt{n}$, and G is an n-vertex graph with*

$$\lambda(G) \;\geqslant\; \lambda(Y_{n,r,q}),$$

then $\#F(G) \geqslant q\,c(n,F)$. Under the above conditions, if G minimizes $\#F(G)$, then $G \in \mathcal{T}_{n,r,q}$.

The bound $q = O(\sqrt{n}\,)$ in Theorem 1.4 is tight up to a constant factor, which reveals a phenomenon different from the edge setting of Theorem 1.1. A spectral condition of this type is not comparable with the edge condition of Theorem 1.1: a graph can satisfy $\lambda(G) \geqslant \lambda(Y_{n,r,q})$ while having fewer than $e(Y_{n,r,q})$ edges. Since a small number of high-degree vertices forces a large spectral radius, the spectral supersaturation statement does not follow from its edge counterpart, and this is closely related to why the range of q shrinks from order n to order $\sqrt{n}$.

Let $\delta_{\mathrm{spec}}(F)$ be the *largest constant* δ_F such that Theorem 1.4 holds. Despite the aforementioned results, no exact value of $\delta_{\mathrm{spec}}(F)$ has been determined in the spectral setting for any color-critical graph F. Unlike the edge setting, which comes from a simple count of edges, the spectral condition $\lambda(G) \geqslant \lambda(Y_{n,r,q})$ works in a different way, as adding q disjoint edges increases the spectral radius by nearly $2q/n$. So the crucial comparison occurs at the second-order term of the spectral expansion of $\lambda(Y_{n,r,q})$. This also explains why the optimal range of q in the spectral setting drops from about n to about $\sqrt{n}$, and why the optimal constant is not a simple fraction. Consequently, determining the exact value of $\delta_{\mathrm{spec}}(F)$ appears more subtle and difficult, and it requires precise control of the Perron vector, instead of the rough estimates that are enough for the order of magnitude.

1.2 Main results

The main contribution of this paper is to establish spectral supersaturation results by determining the exact thresholds $\delta_{\mathrm{spec}}(K_{r+1}) = (1-\frac{1}{r})\sqrt{2}$ and $\delta_{\mathrm{spec}}(C_{2k+1}) = \frac{1}{\sqrt{2}}$; see Table 1. Our results can be viewed as the spectral counterparts of Theorems 1.2 and 1.3. We present these results in the following precise statement. For a graph F with $\chi(F) = r+1$, we recall that $c(n,F)$ denotes the minimum number of copies of F created by adding a single edge to the Turán graph $T_{n,r}$.

Theorem 1.5. *For every $r \geqslant 2$, write $\delta_r := (1-\frac{1}{r})\sqrt{2}$. For every $\varepsilon > 0$, if n is sufficiently large and $1 \leqslant q \leqslant (\delta_r - \varepsilon)\sqrt{n}$, then every n-vertex graph G with $\lambda(G) \geqslant \lambda(Y_{n,r,q})$ satisfies*

$$\#K_{r+1}(G) \;\geqslant\; q\,c(n,K_{r+1}),$$

with equality only if $G \in \mathcal{T}_{n,r,q}$. Moreover, the constant δ_r is best possible.

We next turn to another typical family of color-critical graphs, the odd cycle C_{2k+1} for all $k \geqslant 1$. For this family of graphs, the threshold is an absolute constant $1/\sqrt{2}$, which is independent of the length $2k+1$, in contrast to the clique threshold δ_r above, which depends on r.

Theorem 1.6. *For every $k \geqslant 1$ and $\varepsilon > 0$, write $\delta_{\mathrm{opt}} := 1/\sqrt{2}$. If n is sufficiently large and $1 \leqslant q \leqslant (\delta_{\mathrm{opt}} - \varepsilon)\sqrt{n}$, then every n-vertex graph G with $\lambda(G) \geqslant \lambda(Y_{n,2,q})$ satisfies*

$$\#C_{2k+1}(G) \;\geqslant\; q\, c(n, C_{2k+1}),$$

with equality only if $G \in \mathcal{T}_{n,2,q}$. Moreover, the constant δ_{opt} is best possible.

Up to now, we summarize the spectral supersaturation results in Table 1.

	Edge setting	Spectral setting
Assumption	$e(G) \;\geqslant\; e(T_{n,r}) + q$	$\lambda(G) \;\geqslant\; \lambda(Y_{n,r,q})$
Range of q	$q \leqslant \delta_F\, n$ (Mubayi [27])	$q \leqslant \delta_F\sqrt{n}$ (Fang–Li–Lin–Ma [12])
Conclusion	$\#F(G) \;\geqslant\; q\, c(n, F)$	$\#F(G) \;\geqslant\; q\, c(n, F)$
Threshold for K_{r+1}	$\frac{1}{r}$ (Lovász–Simonovits [24])	$\big(1 - \frac{1}{r}\big)\sqrt{2}$ (Theorem 1.5)
Threshold for C_{2k+1}	$\frac{1}{2}$ (Pikhurko–Yilma [36])	$\frac{1}{\sqrt{2}}$ (Theorem 1.6)

Table 1: Supersaturation in the edge setting versus the spectral setting.

In contrast with the edge setting, Theorems 1.5 and 1.6 differ from the edge setting in two ways: the scale of q and the value of the threshold. First, the conclusion of Theorem 1.1 holds for all $q \leqslant \delta_F n$, of order n, while the range in the spectral setting is of order $\sqrt{n}$. Second, in Theorems 1.2 and 1.3, we see that $\delta_{\mathrm{edge}}(K_{r+1}) = 1/r$ and $\delta_{\mathrm{edge}}(C_{2k+1}) = 1/2$, whereas our spectral thresholds are $\delta_{\mathrm{spec}}(K_{r+1}) = (1-\frac{1}{r})\sqrt{2}$ and $\delta_{\mathrm{spec}}(C_{2k+1}) = 1/\sqrt{2}$. So the spectral threshold must be determined by a different approach. The two settings behave in opposite directions as r grows: the edge threshold $1/r$ decreases to 0, while the spectral threshold $(1 - \frac{1}{r})\sqrt{2}$ increases to $\sqrt{2}$. In the edge setting, $\delta_{\mathrm{edge}}(F)$ depends on the fine structure of F, not only on its chromatic number: it equals $1/r$ for K_{r+1}, but $(r-1)/r^2$ for $K_{r+2} - e$. We speculate that the spectral threshold $\delta_{\mathrm{spec}}(F)$ may depend only on the chromatic number of F, if true, this would be a new feature with no edge analogue.

The scale $\sqrt{n}$ has a transparent spectral meaning. Adding q edges to Turán's graph $T_{n,r}$ raises the spectral radius by $2q/n$ at first order, regardless of how the added edges are distributed. *The distribution becomes visible only at second order term*, where a star gains an extra $q^2(1 - \frac{1}{r})^{-1}/n^2$, while an unbalanced pair of partite sets loses $2(1 - \frac{1}{r})/n$. These two effects balance when q has order $\sqrt{n}$, and the constant $(1 - \frac{1}{r})\sqrt{2}$ records the point of balance. Thus, any first-order estimate is too coarse to detect the threshold of q, which is determined at the second-order estimate.

The supersaturation result in Theorem 1.4 compares G with the candidate graph $Y_{n,r,q}$ and lets q range up to $O(\sqrt{n})$. A more refined goal of this paper is to bound $\#F(G)$ directly by the increment $\lambda(G) - \lambda(T_{n,r})$. Adding q disjoint edges raises the spectral radius by about $2q/n$, so the range $q = O(\sqrt{n})$ corresponds to an increment of order $n^{-1/2}$. As an application of our approach, we establish the following continuous version of Theorem 1.4.

Theorem 1.7. *For any color-critical graph F with $\chi(F) = r+1$, there exists $\delta_F > 0$ such that for all sufficiently large n, and all real C with $0 < C \leqslant \delta_F n^{-1/2}$, if G is an n-vertex graph with*

$$\lambda(G) \geqslant \lambda(T_{n,r}) + C,$$

then

$$\#F(G) \geqslant \Big(\frac{C}{2} - o(1)\Big) n \cdot c(n, F).$$

Moreover, this bound is tight, as witnessed by the graph $Y_{n,r,q}$ for all small q.

Remark 1. Taking $F = K_{r+1}$, Theorem 1.7 improves the Bollobás–Nikiforov bound (1) in a small range above $\lambda(T_{n,r})$. The latter states that if $\lambda(G) = \lambda(T_{n,r}) + \beta n$ for some $\beta > 0$, then

$$\#K_{r+1}(G) \geqslant \beta \frac{r(r-1)}{r+1}\Big(\frac{n}{r}\Big)^{r+1} = \beta n \frac{r-1}{r(r+1)} \cdot n\Big(\frac{n}{r}\Big)^{r-1}.$$

Both bounds are linear in the spectral increment $\lambda(G) - \lambda(T_{n,r})$. Take $\beta \leqslant \delta n^{-3/2}$ with a constant $0 < \delta \leqslant \delta_F$, so that the spectral increment is at most $\delta n^{-1/2}$ and Theorem 1.7 applies. By definition, we have $c(n, K_{r+1}) = (1 + o(1))(\frac{n}{r})^{r-1}$. Consequently, our bound in Theorem 1.7 implies

$$\#K_{r+1}(G) \geqslant \Big(\frac{\delta}{2} - o(1)\Big)\sqrt{n}\,\Big(\frac{n}{r}\Big)^{r-1}.$$

In contrast, the Bollobás–Nikiforov bound gives

$$\#K_{r+1}(G) \geqslant \frac{(r-1)\delta}{r(r+1)}\sqrt{n}\,\Big(\frac{n}{r}\Big)^{r-1}.$$

The former exceeds the latter by a factor $\frac{r(r+1)}{2(r-1)}$, which is linearly large in terms of r. In particular, in the range $0 < \lambda(G) - \frac{n}{2} \leqslant \delta n^{-1/2}$ for some $\delta > 0$, we get $\#K_3(G) \geqslant \big(\frac{1}{4} - o(1)\big)n^2\big(\lambda(G) - \frac{n}{2}\big)$.

Counting cliques above the classical Turán threshold has a long history in the edge setting, from Erdős [8, 9], Lovász–Simonovits [24] and Razborov [37] to Liu–Pikhurko–Staden [20] for triangles, Nikiforov [32] for K_4, and Reiher [38] for all cliques. Theorem 1.7 can be viewed as a spectral counterpart in a narrow range just above $\lambda(T_{n,r})$, where the number of cliques grows linearly in the spectral increment $\lambda(G) - \lambda(T_{n,r})$. For a general graph F, one expects to count the number of copies of F in any graph G with $\lambda(G) \geqslant \lambda(T_{n,r}) + C$ for any universal constant $C > 0$.

1.3 Proof overview and organization

Throughout the paper, we assume that r is fixed and n is sufficiently large, and all constants hidden in $O_r(\cdot)$ may depend on r, but not on n. Our proofs of Theorems 1.5 and 1.6 are based on the spectral supersaturation-stability method. First, we reduce any extremal graph to one that is close to a nearly balanced complete r-partite graph. After the stability reduction, the graph differs from a nearly balanced complete r-partite graph by only $O(\sqrt{n}\,)$ edge changes. Next, a key ingredient is the following second-order estimate on the spectral radius.

Theorem 1.8 (The second-order estimate)**.** *For any $r \geqslant 2$, write $\alpha := \frac{r-1}{r}$, let $K = K_r(n_1, \dots, n_r)$ be an n-vertex complete r-partite graph, and let G be obtained from K by adding s class-edges and deleting h cross-edges, where $s, h = O_r(\sqrt{n})$. Assume $D := e(T_{n,r}) - e(K) = O_r(\sqrt{n})$. Then*

$$\lambda(G) \leqslant \lambda(T_{n,r}) - \frac{2\alpha D}{n} + \frac{2(s-h)}{n} + \frac{(s^2+h^2)\alpha^{-1}}{n^2} + O_r(n^{-5/4}).$$

This second-order estimate controls the spectral radius of a complete r-partite graph after $O(\sqrt{n})$ class-edge additions and cross-edge deletions, which appears in many spectral stability arguments and may be useful for other spectral extremal problems where first-order estimates are too coarse.

Proof overview. For Theorem 1.5, we apply a structural result of Fang, Li, Lin and Ma (Theorem 2.1), which transforms any extremal graph satisfying $\lambda(G) \geqslant \lambda(Y_{n,r,q})$ and $\#K_{r+1}(G) \leqslant q\,c(n, K_{r+1})$ into a graph that is built from a nearly balanced complete r-partite graph K by adding s class-edges and deleting h cross-edges, where $s, h = O_r(\sqrt{n}\,)$ and $D := e(T_{n,r}) - e(K) = O_r(\sqrt{n}\,)$ (Lemma 4.2). For such graphs, we then prove two precise estimates. The first is a second-order expansion of $\lambda(G)$ as stated in Theorem 1.8, which we get by studying the Perron vector of K up to second order. Comparing with $\lambda(Y_{n,r,q}) = \lambda(T_{n,r}) + 2q/n + O_r(n^{-3/2})$ (Lemma 2.4) gives the key inequality:

$$0 \leqslant \lambda(G) - \lambda(Y_{n,r,q}) \leqslant \frac{2(s-q-h)}{n} - \frac{2\alpha D}{n} + \frac{(s^2+h^2)\alpha^{-1}}{n^2} + o(n^{-1}). \tag{2}$$

The second is a counting lemma (Lemma 4.3): each added class-edge of G creates about $c(n, K_{r+1})$ copies of K_{r+1}, up to small losses coming from the imbalance D and the deletions h, that is,

$$\#K_{r+1}(G) \geqslant s\, c(n, K_{r+1}) - C_r\, s \left(\sqrt{D} + h + 1\right) n^{r-2}.$$

Combining with the assumption $\#K_{r+1}(G) \leqslant q\, c(n, K_{r+1})$, we get $\sqrt{D} + h + 1 \geqslant \kappa \frac{(s-q)n}{s}$ for a constant $\kappa = \kappa(r) > 0$. Next, we show that the right-hand side of (2) is negative for $s \neq q$. In the case $s < q$, the spectral estimate (2) gives $\lambda(G) - \lambda(Y_{n,r,q}) \leqslant \frac{2(s-q)}{n} + \frac{s^2\alpha^{-1}}{n^2} + o(n^{-1}) < 0$. In the case $s > q$, we write $s := q + \ell$ for some $\ell \geqslant 1$. Then $\sqrt{D} + h + 1 \geqslant \kappa \frac{\ell n}{s}$, which yields $\frac{2h+2\alpha D}{n} \geqslant \frac{2\alpha\kappa\ell}{s} - \frac{2\alpha}{n}$. So $\lambda(G) - \lambda(Y_{n,r,q}) \leqslant \frac{2\ell}{n} - \frac{2\alpha\kappa\ell}{s} + O_r(n^{-1}) < 0$ since $s = O_r(\sqrt{n})$. Now, we consider the case $s = q$. If $h \geqslant 1$ or $D \geqslant 1$, then one can verify that $\frac{-2h}{n} - \frac{2\alpha D}{n} + \frac{h^2\alpha^{-1}}{n^2} \leqslant -\frac{2\alpha}{n} + o(n^{-1})$. Thus $\lambda(G) - \lambda(Y_{n,r,q}) \leqslant -\frac{2\alpha}{n} + \frac{q^2\alpha^{-1}}{n^2} + o(n^{-1}) < 0$, a contradiction. Consequently, we must have $s = q$ and $h = D = 0$. Hence G is obtained from the Turán graph $T_{n,r}$ by adding q edges.

To see that the constant δ_r cannot be improved, fix $q = (\delta_r + \varepsilon)\sqrt{n}$ and let $H = H_{n,r,q}$ be built from the unbalanced complete r-partite graph $K_r(m+1, m-1, m, \ldots, m)$ (where $n = rm$, with one part enlarged and one shrunk by one vertex) by adding a star with q edges inside the part of size $m+1$. Shrinking a part lowers the count, so H has only $q(m-1)m^{r-2} < q\, c(n, K_{r+1})$ copies of K_{r+1}. Yet the added star raises the spectral radius enough that $\lambda(H) > \lambda(Y_{n,r,q})$ once $q > (\delta_r + \varepsilon)\sqrt{n}$ (Example 4.1). Thus, above the threshold δ_r, the spectral condition $\lambda(H) > \lambda(Y_{n,r,q})$ no longer forces $q\, c(n, K_{r+1})$ copies. The threshold $\delta_r = \sqrt{2\,\alpha}$ is exactly the point at which the second-order term gain $q^2\alpha^{-1}/n^2$ from concentrating the added edges into a star balances the loss $2\alpha/n$ from making the underlying r-partite graph unbalanced, that is, $q^2\alpha^{-1} = 2\alpha n$, so $q = \sqrt{2}\alpha \cdot \sqrt{n}$.

The proof of Theorem 1.6 follows the same plan. The stability input is Lemma 5.2, the second-order spectral estimate is Lemma 5.3, and the counting step is Lemma 5.4, which shows that each added class-edge lies in about $c(n, C_{2k+1})$ copies of C_{2k+1}. The same three cases $s < q$, $s = q$, $s > q$ give the bound, and the construction in Example 5.1 shows that $1/\sqrt{2}$ is best possible.

Organization. In Section 2, we introduce the tools: a second-order expansion for the spectral radius of a complete r-partite graph, the expansion of $\lambda(Y_{n,r,q})$, and a sharp construction obtained by adding a star. In Section 3, we first prove the key estimate in Theorem 1.8, and then we apply it to prove Theorem 1.7. In Section 4, we prove Theorem 1.5, and in Section 5, we prove Theorem 1.6. In Section 6, we conclude with some open spectral extremal problems for interested readers.

2 Preliminaries

We need the following structural theorem of Fang, Li, Lin and Ma [12].

Theorem 2.1 (Fang–Li–Lin–Ma [12]). *Let $0 < \varepsilon_1 < 1/2$, and let F be a color-critical graph with $\chi(F) = r + 1$. There are constants $\delta_F, \eta_F > 0$ such that the following holds for every sufficiently large n. If $1 \leqslant q \leqslant \delta_F n$ and G is an n-vertex graph satisfying*

$$\lambda(G) \geqslant \lambda(T_{n,r})$$

and

$$\#F(G) \leqslant (q + \varepsilon_1)c(n, F),$$

then G can be obtained from a complete r-partite graph $K_r(n_1, n_2, \ldots, n_r)$ by adding s class-edges and deleting h cross-edges, where $h \leqslant 2s \leqslant 4q/\eta_F$ and $|n_i - n_j| \leqslant 4\sqrt{s}$ for all $i \neq j$.

The following lemma bounds the sum of squares of the degrees in a graph of given size.

Lemma 2.2 (See [35]). *If G is a graph with m edges, then*

$$\sum_{v \in V(G)} d^2(v) \leqslant m^2 + m.$$

2.1 The spectral radius of complete multipartite graphs

First, we estimate the spectral radius of a nearly balanced complete r-partite graph.

Theorem 2.3. *Let $K = K_r(n_1, n_2, \dots, n_r)$ be a complete r-partite graph on n vertices. Write $\alpha = \frac{r-1}{r}$ and $n_i = \frac{n}{r} + a_i$, where $\sum_{i=1}^r a_i = 0$, and assume $B := \sum_{i=1}^r a_i^2 = O_r(\sqrt{n})$. Then*

$$\lambda(K) = \alpha n - \frac{\alpha B}{n} + O_r(n^{-5/4}). \tag{3}$$

Moreover, if $D := e(T_{n,r}) - e(K)$, then

$$\lambda(K) = \lambda(T_{n,r}) - \frac{2\alpha D}{n} + O_r(n^{-5/4}).$$

Proof. Denote $\lambda = \lambda(K)$. By the Perron–Frobenius theorem, the Perron vector of K is positive and, by symmetry, constant on each part. Let x_i be its value on the part V_i and $S := \sum_j n_j x_j$. A vertex of part V_i is adjacent to all vertices outside part V_i, so $\lambda x_i = \sum_{j \neq i} n_j x_j = S - n_i x_i$, i.e.,

$$x_i = \frac{S}{\lambda + n_i} > 0.$$

Multiplying by n_i, summing, and dividing by $S > 0$ gives the characteristic equation

$$\sum_{i=1}^r \frac{n_i}{\lambda + n_i} = 1. \tag{4}$$

Put $m := n/r$ and $\lambda := \alpha n + z$ for some z, which will be determined later. Since K is K_{r+1}-free, Wilf's bound gives $\lambda \leqslant \alpha n$. On the other hand, $\lambda \geqslant \frac{2e(K)}{n} = \frac{1}{n}\big(n^2 - \sum_i n_i^2\big) = \alpha n - \frac{B}{n} \geqslant \alpha n - 1$ for large n, since $B = O_r(\sqrt{n})$. Hence the value $z = \lambda - \alpha n$ lies in $[-1, 0]$, and it is therefore the unique root of F there. Since $\alpha n + m = n$, we have $\lambda + n_i = n + z + a_i$, so (4) reads $F(z) = 0$, where

$$F(z) := \sum_{i=1}^r \frac{m + a_i}{n + z + a_i} - 1.$$

Since λ is the largest root of (4), the corresponding z is the root of F.

From $a_i^2 \leqslant B = O_r(\sqrt{n})$ we get $|a_i| \leqslant \sqrt{B} = O_r(n^{1/4})$. Hence for $|z| \leqslant 1$ and all i, we have $|z + a_i| = O_r(n^{1/4})$, so $(n + z + a_i)^2 = n^2\big(1 + O_r(n^{-3/4})\big)$. Since $\sum_i (m + a_i) = n$, we get

$$F'(z) = -\sum_{i=1}^r \frac{m + a_i}{(n + z + a_i)^2} = -\frac{1}{n^2} \sum_{i=1}^r (m + a_i)\big(1 + O_r(n^{-3/4})\big) = -\frac{1}{n} + O_r(n^{-7/4}). \tag{5}$$

In particular $F'(z) \leqslant -\frac{1}{2n}$ on $z \in [-1, 1]$ for large n, so F has a unique root there.

Using $\frac{1}{r} + \alpha = 1$, we have $m + a_i = \frac{1}{r}(n + a_i) + \alpha a_i$. Then $\frac{m + a_i}{n + a_i} = \frac{1}{r} + \frac{\alpha a_i}{n + a_i}$. So

$$F(0) = \alpha \sum_{i=1}^r \frac{a_i}{n + a_i}.$$

From $\frac{a_i}{n + a_i} = \frac{a_i}{n} - \frac{a_i^2}{n^2} + O_r\big(\frac{|a_i|^3}{n^3}\big)$ and $\sum_i a_i = 0$, we get

$$\sum_{i=1}^r \frac{a_i}{n + a_i} = -\frac{B}{n^2} + O_r\Big(\frac{\sum_i |a_i|^3}{n^3}\Big) = -\frac{B}{n^2} + O_r(n^{-9/4}),$$

where $\sum_i |a_i|^3 \leqslant (\max_i |a_i|)\, B = O_r(n^{1/4}) \cdot O_r(\sqrt{n}) = O_r(n^{3/4})$. Therefore

$$F(0) = -\frac{\alpha B}{n^2} + O_r(n^{-9/4}). \tag{6}$$

By (6), we have $|F(0)| = O_r(n^{-3/2})$, so with $F' \leqslant -\frac{1}{2n}$ the root satisfies $|z| \leqslant 2n|F(0)| = O_r(n^{-1/2}) < 1$. By the mean value theorem, we get $F(z) - F(0) = F'(\zeta) \cdot z$ for some ζ between 0 and z. Thus, we get $z = -\frac{F(0)}{F'(\zeta)}$. Inserting (5) and (6),

$$z = -\frac{-\alpha B/n^2 + O_r(n^{-9/4})}{-\frac{1}{n} + O_r(n^{-7/4})} = -\frac{\alpha B}{n} + O_r(n^{-5/4}).$$

Hence, we obtain $\lambda(K) = \alpha n + z = \alpha n - \frac{\alpha B}{n} + O_r(n^{-5/4})$, which proves (3).

We write the part sizes of the Turán graph $T_{n,r}$ as $t_i = n/r + b_i$ for $i \in \{1, \dots, r\}$, where $\sum_i b_i = 0$. Since $T_{n,r}$ is balanced, i.e., $|t_i - t_j| \leqslant 1$ for all $i \neq j$, we have $|b_i| \leqslant 1$ and

$$B_T := \sum_{i=1}^{r} b_i^2 = O_r(1).$$

Applying (3) to $T_{n,r}$ gives

$$\lambda(T_{n,r}) = \alpha n - \frac{\alpha B_T}{n} + O_r(n^{-5/4}). \tag{7}$$

For any complete r-partite graph with part sizes $x_1, \dots, x_r$ summing to n, we have

$$e(K_r(x_1, \dots, x_r)) = \frac{1}{2}\left(n^2 - \sum_{i=1}^{r} x_i^2\right).$$

Since $\sum_i n_i^2 = \sum_i (\frac{n}{r} + a_i)^2 = \frac{n^2}{r} + B$ and $\sum_i t_i^2 = \frac{n^2}{r} + B_T$, we obtain

$$D = e(T_{n,r}) - e(K) = \frac{1}{2}\left(\sum_i n_i^2 - \sum_i t_i^2\right) = \frac{1}{2}(B - B_T). \tag{8}$$

Subtracting (7) from (3), it follows that

$$\lambda(K) = \lambda(T_{n,r}) - \frac{\alpha(B - B_T)}{n} + O_r(n^{-5/4}) = \lambda(T_{n,r}) - \frac{2\alpha D}{n} + O_r(n^{-5/4}). \qquad \square$$

2.2 The spectral radius of Turán's graph plus a matching

The graph $Y_{n,r,q}$ is obtained from the Turán graph $T_{n,r}$ by adding a matching with q edges into a largest part. We show that such a matching raises the spectral radius by $2q/n + O_r(n^{-3/2})$.

Lemma 2.4. *If q is a positive integer with $q = O_r(\sqrt{n})$, then*

$$\lambda(Y_{n,r,q}) = \lambda(T_{n,r}) + \frac{2q}{n} + O_r(n^{-3/2}).$$

Proof. Let A_T be the adjacency matrix of the Turán graph $T_{n,r}$. Let M be the adjacency matrix of the q disjoint edges added to $T_{n,r}$ to form $Y_{n,r,q}$. Then

$$A(Y_{n,r,q}) = A_T + M.$$

Let $\mu := \lambda(T_{n,r})$ and let $\boldsymbol{u} \in \mathbb{R}^n$ be a unit Perron vector of the Turán graph $T_{n,r}$. Because the part sizes of $T_{n,r}$ differ by at most one, we first show that for every vertex v,

$$u_v = \frac{1}{\sqrt{n}} + O_r(n^{-3/2}). \tag{9}$$

By the Perron–Frobenius theorem and the symmetry of $T_{n,r}$, the coordinate u_v is constant on each part. We write $u_v = c_i$ for v in part i. The eigenvalue equation at a vertex of part i reads $\mu c_i = \sum_{j \neq i} t_j c_j = C - t_i c_i$, where $C := \sum_j t_j c_j$. Hence, we get

$$c_i = \frac{C}{\mu + t_i}. \tag{10}$$

The part sizes of T differ by at most one and satisfy $t_i = n/r + O_r(1)$ for all i. Since $\delta(T_{n,r}) \leqslant \mu \leqslant \Delta(T_{n,r})$, we have $\mu = \alpha n + O_r(1)$ and $\mu + t_i = n + O_r(1)$. Then $\frac{1}{\mu+t_i} = \frac{1}{n}\big(1 + O_r(n^{-1})\big)$. Using normalization and $\sum_i t_i = n$ gives

$$1 = \sum_v u_v^2 = \sum_{i=1}^{r} t_i c_i^2 = \sum_{i=1}^{r} \frac{t_i C^2}{(\mu + t_i)^2} = \frac{C^2}{n^2}\left(\sum_{i=1}^{r} t_i\right)\big(1 + O_r(n^{-1})\big) = \frac{C^2}{n}\big(1 + O_r(n^{-1})\big).$$

Therefore $C^2 = n + O_r(1)$, i.e., $C = \sqrt{n} + O_r(n^{-1/2})$. Then

$$c_i = \frac{C}{\mu + t_i} = \frac{\sqrt{n} + O_r(n^{-1/2})}{n + O_r(1)} = \frac{1}{\sqrt{n}} + O_r(n^{-3/2}),$$

which proves the estimate in (9).

First we prove the lower bound on $\lambda(Y_{n,r,q})$. By the Rayleigh quotient,

$$\lambda(Y_{n,r,q}) \geqslant \boldsymbol{u}^T(A_T + M)\boldsymbol{u} = \mu + \boldsymbol{u}^T M \boldsymbol{u}.$$

Each added matching edge xy contributes $2u_x u_y$ to $\boldsymbol{u}^T M \boldsymbol{u}$. By (9),

$$2u_x u_y = \frac{2}{n} + O_r(n^{-2}).$$

There are q added edges, so

$$\boldsymbol{u}^T M \boldsymbol{u} = \frac{2q}{n} + O_r\left(\frac{q}{n^2}\right) = \frac{2q}{n} + O_r(n^{-3/2}). \tag{11}$$

Thus

$$\lambda(Y_{n,r,q}) \geqslant \mu + \frac{2q}{n} + O_r(n^{-3/2}). \tag{12}$$

Now we prove the upper bound on $\lambda(Y_{n,r,q})$. Let $\boldsymbol{x} \in \mathbb{R}^n$ be any unit vector. Write

$$\boldsymbol{x} = a\boldsymbol{u} + \boldsymbol{y}, \qquad \boldsymbol{y} \perp \boldsymbol{u}, \qquad a^2 + \|\boldsymbol{y}\|^2 = 1.$$

Put $\beta := \|\boldsymbol{y}\|$. Since $T_{n,r}$ is a complete r-partite graph, $T_{n,r}$ has exactly one positive eigenvalue. Therefore $\boldsymbol{y}^T A_T \boldsymbol{y} \leqslant 0$. It follows that $\boldsymbol{x}^T A_T \boldsymbol{x} \leqslant \mu a^2$. Also,

$$\boldsymbol{x}^T M \boldsymbol{x} = a^2 \boldsymbol{u}^T M \boldsymbol{u} + 2a\, \boldsymbol{y}^T M \boldsymbol{u} + \boldsymbol{y}^T M \boldsymbol{y}.$$

Since M is the adjacency matrix corresponding to the added matching, we get $\|M\| = 1$. Hence

$$\boldsymbol{y}^T M \boldsymbol{y} \leqslant \|M\|\|\boldsymbol{y}\|^2 = \beta^2.$$

Using $a^2 = 1 - \beta^2$ and $|a| \leqslant 1$, we obtain

$$\begin{aligned}\boldsymbol{x}^T(A_T + M)\boldsymbol{x} &\leqslant \mu a^2 + a^2 \boldsymbol{u}^T M \boldsymbol{u} + 2|a|\,|\boldsymbol{y}^T M \boldsymbol{u}| + \beta^2 \\ &\leqslant \mu + \boldsymbol{u}^T M \boldsymbol{u} - (\mu + \boldsymbol{u}^T M \boldsymbol{u} - 1)\beta^2 + 2\beta\|M\boldsymbol{u}\|.\end{aligned}$$

Since $q = O_r(\sqrt{n})$, we have $\boldsymbol{u}^T M \boldsymbol{u} = o(1)$ by (11). For sufficiently large n, we get

$$\mu + \boldsymbol{u}^T M \boldsymbol{u} - 1 = \alpha n + o(n) > 0.$$

For $L > 0$, the maximum of $-L\beta^2 + 2B\beta$ over $\beta \geqslant 0$ is B^2/L. Applying this with

$$L := \mu + \boldsymbol{u}^T M \boldsymbol{u} - 1, \qquad B := \|M\boldsymbol{u}\|,$$

we get

$$\lambda(Y_{n,r,q}) \leqslant \mu + \boldsymbol{u}^T M \boldsymbol{u} + \frac{\|M\boldsymbol{u}\|^2}{\mu + \boldsymbol{u}^T M \boldsymbol{u} - 1}. \tag{13}$$

It remains to estimate the last term. Because M is a matching, $M\boldsymbol{u}$ is supported on the $2q$ endpoints of the matching, and each nonzero coordinate is $1/\sqrt{n} + O_r(n^{-3/2})$. Therefore

$$\|M\boldsymbol{u}\|^2 = 2q\left(\frac{1}{n} + O_r(n^{-2})\right) = O_r\left(\frac{q}{n}\right) = O_r(n^{-1/2}).$$

The denominator in (13) is $\alpha n + o(n)$, so

$$\frac{\|M\boldsymbol{u}\|^2}{\mu + \boldsymbol{u}^T M\boldsymbol{u} - 1} = O_r(n^{-3/2}).$$

Combining this with (11) and (13) gives

$$\lambda(Y_{n,r,q}) \leqslant \mu + \frac{2q}{n} + O_r(n^{-3/2}).$$

Together with the lower bound (12), this proves the lemma. □

2.3 A sharp construction by adding a star

To show the tightness of δ_r in Theorem 1.5, we need to construct extremal examples with spectral radius at least $\lambda(Y_{n,r,q})$ for infinitely many n. Throughout this section, we assume that $n = rm$.

Proposition 2.5. *Assume that $r \geqslant 2$ and $\alpha = \frac{r-1}{r}$. Suppose that $n = rm$ is sufficiently large and $q = O_r(\sqrt{n})$. Let $H = H_{n,r,q}$ be the graph obtained from the complete r-partite graph $K_0 := K_r(m+1, m-1, m, \ldots, m)$ by adding a q-edge star into the partite set of size $m+1$. Then*

$$\lambda(H) \geqslant \alpha n - \frac{2\alpha}{n} + \frac{2q}{n} + \frac{q^2\alpha^{-1}}{n^2} + O_r(n^{-3/2}).$$

By Theorem 2.3, we can get $\lambda(K_0) \approx \alpha n - \frac{2\alpha}{n}$. The remaining argument is to prove that adding a star with q edges increases $\lambda(K_0)$ by about $\frac{2q}{n} + \frac{q^2\alpha^{-1}}{n^2}$. Here, the second order term is $\frac{q^2\alpha^{-1}}{n^2}$, which is different from that of adding a matching. To determine this term, we use a test vector spanned by $\boldsymbol{u}$ and a second vector $\boldsymbol{y}$, which we get by centering the star-degree function on the part that contains the star. This $\boldsymbol{y}$ is orthogonal to $\boldsymbol{u}$ and is sent to zero by $A(K_0)$, but $\boldsymbol{y}^T S\boldsymbol{u}$ is large because of the star; choosing the best vector in this plane then gives this missing term exactly.

Proof. First, we apply Theorem 2.3 to estimate the spectral radius of K_0. Comparing to the balanced $K_r(m, m, \ldots, m)$, the deviations are $1, -1, 0, \ldots, 0$. Thus $B = 1^2 + (-1)^2 = 2$. Since the deviations of K_0 are bounded ($\max_i |a_i| = 1$ and $B = 2$), the estimates in the proof of Theorem 2.3 sharpen to $F(0) = -\frac{\alpha B}{n^2} + O_r(n^{-3})$ and $F'(z) = -\frac{1}{n} + O_r(n^{-2})$. Hence

$$\lambda(K_0) = \alpha n - \frac{2\alpha}{n} + O_r(n^{-2}).$$

Let S be the adjacency matrix of the added q-edge star. Let $\boldsymbol{u}$ be the unit Perron vector of K_0. Since the parts of K_0 differ from n/r by only $O_r(1)$, as in (9), the coordinates of the Perron vector $\boldsymbol{u}$ are almost equal, and they are given as

$$u_0 = \frac{1}{\sqrt{n}} + O_r(n^{-3/2}). \tag{14}$$

The first-order term of the added star is

$$\boldsymbol{u}^T S\boldsymbol{u} = 2qu_0^2 = \frac{2q}{n} + O_r\left(\frac{q}{n^2}\right) = \frac{2q}{n} + O_r(n^{-3/2}). \tag{15}$$

From the above discussion, the Rayleigh quotient implies $\lambda(H) \geqslant \alpha n - 2\alpha/n + 2q/n + O_r(n^{-3/2})$. Thus, the difficulty is to show the second-order term $q^2\alpha^{-1}/n^2$ of the increment caused by the added star. To achieve this, we need to construct a new test vector instead of $\boldsymbol{u}$.

Let U be the partite set in K_0 of size $|U| = m+1$, let c be the center vertex of the star, and let L be the set of its q leaves. Define the star-degree function d on U by $d(c) = q$, $d(x) = 1$ for $x \in L$, and $d(x) = 0$ for $x \in U \setminus (\{c\} \cup L)$. Then $\sum_{x\in U} d(x) = 2q$. Let

$$\bar{d} := \frac{2q}{|U|} = \frac{2q}{m+1}.$$

Define a vector $\boldsymbol{w} \in \mathbb{R}^n$ by

$$w_x = \begin{cases} d(x) - \bar{d}, & \text{if } x \in U; \\ 0, & \text{if } x \notin U. \end{cases}$$

Then

$$\sum_{x \in U} w_x = \sum_{x \in U} d(x) - |U|\bar{d} = 2q - (m+1)\frac{2q}{m+1} = 0.$$

Because $\boldsymbol{u}$ is constant restricted to U and $\boldsymbol{w}$ is supported only on U, this gives $\boldsymbol{w} \perp \boldsymbol{u}$. Since $\boldsymbol{w}$ is supported on one part and has coordinate sum zero within that part, the adjacency matrix $A(K_0)$ sends $\boldsymbol{w}$ to zero, that is, $A(K_0)\boldsymbol{w} = 0$. We next compute $\|\boldsymbol{w}\|^2$. Since $w_x = d(x) - \bar{d}$ on U,

$$\|\boldsymbol{w}\|^2 = \sum_{x \in U} (d(x) - \bar{d})^2 = \sum_{x \in U} d(x)^2 - 2\bar{d} \sum_{x \in U} d(x) + |U|\bar{d}^2.$$

Now

$$\sum_{x \in U} d(x)^2 = q^2 + q,$$

while

$$2\bar{d} \sum_{x \in U} d(x) = 2 \cdot \frac{2q}{m+1} \cdot 2q = \frac{8q^2}{m+1},$$

and

$$|U|\bar{d}^2 = (m+1)\frac{4q^2}{(m+1)^2} = \frac{4q^2}{m+1}.$$

Since $q = O(\sqrt{n})$, we get

$$\|\boldsymbol{w}\|^2 = q^2 + q - \frac{4q^2}{m+1} = q^2 + O(q). \tag{16}$$

Let $\boldsymbol{y} := \frac{\boldsymbol{w}}{\|\boldsymbol{w}\|}$. Since $\boldsymbol{w} \perp \boldsymbol{u}$ and $A(K_0)\boldsymbol{w} = 0$, we get

$$\|\boldsymbol{y}\| = 1, \qquad \boldsymbol{y} \perp \boldsymbol{u}, \qquad A(K_0)\boldsymbol{y} = 0. \tag{17}$$

We compute $\boldsymbol{y}^T S \boldsymbol{u}$. Since S is the adjacency matrix of the star and $u_x = u_0$ for all $x \in U$, we have $(S\boldsymbol{u})_x = u_0\, d(x)$ if $x \in U$; and $(S\boldsymbol{u})_x = 0$ if $x \notin U$. Hence

$$\boldsymbol{w}^T S \boldsymbol{u} = u_0 \sum_{x \in U} w_x d(x).$$

But

$$\sum_{x \in U} w_x d(x) = \sum_{x \in U} d(x)^2 - \bar{d} \sum_{x \in U} d(x) = (q^2 + q) - \frac{2q}{m+1} \cdot 2q = q^2 + q - \frac{4q^2}{m+1}.$$

Therefore $\boldsymbol{w}^T S \boldsymbol{u} = u_0 \|\boldsymbol{w}\|^2$ by (16), and so

$$\boldsymbol{y}^T S \boldsymbol{u} = u_0 \|\boldsymbol{w}\|. \tag{18}$$

Let $\mu := \lambda(K_0)$ and $t := \boldsymbol{y}^T S \boldsymbol{u}/\mu$. Then the vector

$$\boldsymbol{x} := \frac{\boldsymbol{u} + t\boldsymbol{y}}{\sqrt{1+t^2}}$$

has norm one. Therefore, by the Rayleigh quotient,

$$\lambda(H) = \lambda(K_0 + S) \geqslant \boldsymbol{x}^T (A(K_0) + S)\boldsymbol{x} = \frac{\mu + \boldsymbol{u}^T S \boldsymbol{u} + 2t\, \boldsymbol{y}^T S \boldsymbol{u} + t^2 \boldsymbol{y}^T S \boldsymbol{y}}{1+t^2}, \tag{19}$$

where we used (17) to remove all cross terms involving $A(K_0)$.

Next, by (14), (16) and (18),

$$(\boldsymbol{y}^T S\boldsymbol{u})^2 = u_0^2\|\boldsymbol{w}\|^2 = \left(\frac{1}{n} + O_r(n^{-2})\right)(q^2 + O(q)) = \frac{q^2}{n} + O_r(n^{-1/2}).$$

Since $\mu = \alpha n + O_r(n^{-1})$, we get

$$\frac{(\boldsymbol{y}^T S\boldsymbol{u})^2}{\mu} = \frac{q^2/n + O_r(n^{-1/2})}{\alpha n + O_r(n^{-1})} = \frac{1}{\alpha}\frac{q^2}{n^2} + O_r(n^{-3/2}). \tag{20}$$

We now simplify the right-hand side of (19). The added star is a copy of $K_{1,q}$, so its adjacency matrix has spectral norm $\|S\| = \sqrt{q} = O_r(n^{1/4})$, and hence $|\boldsymbol{y}^T S\boldsymbol{y}| \leqslant \|S\|\,\|\boldsymbol{y}\|^2 = O_r(n^{1/4})$. By (18), (14) and (16), we have $\boldsymbol{y}^T S\boldsymbol{u} = u_0\|\boldsymbol{w}\| = O_r(1)$. Since $\mu = \alpha n + O_r(n^{-1})$, we get $t = \boldsymbol{y}^T S\boldsymbol{u}/\mu = O_r(n^{-1})$. Note that $t\,\boldsymbol{y}^T S\boldsymbol{u} = (\boldsymbol{y}^T S\boldsymbol{u})^2/\mu = \mu t^2$. So the numerator in (19) regroups as

$$\mu + \boldsymbol{u}^T S\boldsymbol{u} + 2t\,\boldsymbol{y}^T S\boldsymbol{u} + t^2\boldsymbol{y}^T S\boldsymbol{y} = \mu(1+t^2) + \boldsymbol{u}^T S\boldsymbol{u} + \frac{(\boldsymbol{y}^T S\boldsymbol{u})^2}{\mu} + t^2\boldsymbol{y}^T S\boldsymbol{y}.$$

Dividing by $1 + t^2$, we find that (19) becomes

$$\frac{\mu + \boldsymbol{u}^T S\boldsymbol{u} + 2t\,\boldsymbol{y}^T S\boldsymbol{u} + t^2\boldsymbol{y}^T S\boldsymbol{y}}{1+t^2} = \mu + \frac{1}{1+t^2}\left(\boldsymbol{u}^T S\boldsymbol{u} + \frac{(\boldsymbol{y}^T S\boldsymbol{u})^2}{\mu} + t^2\boldsymbol{y}^T S\boldsymbol{y}\right). \tag{21}$$

By (15) and (20), we get $\boldsymbol{u}^T S\boldsymbol{u} = O_r(n^{-1/2})$ and $(\boldsymbol{y}^T S\boldsymbol{u})^2/\mu = O_r(n^{-1})$, while

$$|t^2\boldsymbol{y}^T S\boldsymbol{y}| \leqslant t^2\|S\| = O_r(n^{-2}) \cdot O_r(n^{1/4}) = O_r(n^{-7/4}).$$

Hence the parenthesis equals $\boldsymbol{u}^T S\boldsymbol{u} + (\boldsymbol{y}^T S\boldsymbol{u})^2/\mu + O_r(n^{-7/4}) = O_r(n^{-1/2})$, and multiplying by $\frac{1}{1+t^2} = 1 + O_r(n^{-2})$ alters it by only $O_r(n^{-1/2}) \cdot O_r(n^{-2}) = O_r(n^{-5/2})$. Therefore (21) gives

$$\lambda(H) \geqslant \mu + \boldsymbol{u}^T S\boldsymbol{u} + \frac{(\boldsymbol{y}^T S\boldsymbol{u})^2}{\mu} + O_r(n^{-7/4}).$$

Using (15) and (20), we obtain

$$\lambda(H) \geqslant \mu + \frac{2q}{n} + \frac{q^2\alpha^{-1}}{n^2} + O_r(n^{-3/2}).$$

Recalling that $\mu = \lambda(K_0) = \alpha n - 2\alpha/n + O_r(n^{-2})$, we obtain the desired bound. □

3 Spectral estimates for nearly Turán graphs

In this section, we first prove the second-order spectral estimate in Theorem 1.8. Second, we prove the continuous supersaturation in Theorem 1.7 by using Theorem 1.8 and Theorem 1.4.

3.1 Proof of Theorem 1.8

Our goal is to bound the Rayleigh quotient $\boldsymbol{x}^T A(G)\boldsymbol{x}$ for an arbitrary unit vector $\boldsymbol{x}$. We write $A(G) := A(K) + E$, where E has a $+1$ for each added class-edge and a -1 for each deleted cross-edge. Note that $A(K)$ has only one positive eigenvalue μ with the unit Perron vector $\boldsymbol{u}$. We denote $\boldsymbol{x} := a\boldsymbol{u} + \boldsymbol{y}$ for some $\boldsymbol{y}$ with $\boldsymbol{y} \perp \boldsymbol{u}$, then bound the part from $A(K)$ by μa^2 and split the rest into a first-order term $e_0 := \boldsymbol{u}^T E\boldsymbol{u}$ and a second-order term coming from $\|E\boldsymbol{u}\|^2/\mu$. The first term gives $2(s-h)/n$, and the second, after we bound $\|E\boldsymbol{u}\|^2$ by the degree-square sum (Lemma 2.2), gives $(s^2+h^2)\alpha^{-1}/n^2$, where the coefficient α^{-1} comes from the spectral radius $\mu \approx \alpha n$.

Proof of Theorem 1.8. As stated above, we denote $A(G) := A_0 + E$, where $A_0 = A(K)$ and E is a symmetric matrix with entries $+1$ corresponding to the added class-edges and -1 corresponding to

the deleted cross-edges. Let $\mu = \lambda(K)$ and let $\boldsymbol{u}$ be the positive unit Perron vector of K. By the identity (8) and $B_T = O_r(1)$, the hypothesis $D = O_r(\sqrt{n})$ gives

$$B := \sum_{i=1}^{r} \Big(n_i - \frac{n}{r}\Big)^2 = O_r(\sqrt{n}),$$

and then $\max_{i\in[r]} |n_i - \frac{n}{r}| = O_r(n^{1/4})$. By Theorem 2.3, we have

$$\mu = \alpha n - \frac{\alpha B}{n} + O_r(n^{-5/4}),$$

which together with $B = O_r(\sqrt{n})$ yields

$$\mu = \alpha n + O_r(n^{-1/2}). \tag{22}$$

Similar to (10), we obtain $u_v = \frac{C}{\mu+n_i}$ for every $v \in V_i$ with $C > 0$. Since $n_i = \frac{n}{r} + O_r(n^{1/4})$, we have $\mu + n_i = n + O_r(n^{1/4})$ and $u_v = \frac{C}{\mu+n_i} = \frac{C}{n}\big(1 + O_r(n^{-3/4})\big)$. Then $\sum_v u_v^2 = 1$ forces

$$C = \sqrt{n}\,\big(1 + O_r(n^{-3/4})\big).$$

Then for every $v \in V$, we have

$$u_v = \frac{1}{\sqrt{n}}\big(1 + O_r(n^{-3/4})\big). \tag{23}$$

Put $e_0 := \boldsymbol{u}^T E \boldsymbol{u}$. An added class-edge contributes $2u_x u_y$ and a deleted cross-edge $-2u_x u_y$. By (23), we have $2u_x u_y = \frac{2}{n} + O_r(n^{-7/4})$. As there are s added and h deleted edges and $s + h = O_r(\sqrt{n})$, we conclude that

$$e_0 = \frac{2(s-h)}{n} + O_r\big((s+h)n^{-7/4}\big) = \frac{2(s-h)}{n} + O_r(n^{-5/4}). \tag{24}$$

By the definition of E, it follows that

$$(E\boldsymbol{u})_v = \sum_{vw \text{ added}} u_w - \sum_{vw \text{ deleted}} u_w.$$

For a vertex v, let $d^+(v)$ and $d^-(v)$ be its degrees in the graphs formed by the added class-edges and by the deleted cross-edges, respectively. By (23), we obtain

$$(E\boldsymbol{u})_v = \frac{1}{\sqrt{n}}\big(d^+(v) - d^-(v)\big) + O_r\big(n^{-5/4}(d^+(v) + d^-(v))\big).$$

We now bound $\|E\boldsymbol{u}\|^2$. Set $X := \sum_v \big(d^+(v)^2 + d^-(v)^2\big)$. Squaring and summing the above over all v, the main term is at most $\frac{1}{n}\sum_v (d^+(v) - d^-(v))^2 \leqslant \frac{1}{n}X$, while the squared error term is $O_r(n^{-5/2})X$, and the cross term is $O_r(n^{-7/4})X$. Hence $\|E\boldsymbol{u}\|^2 \leqslant \frac{1}{n}X + O_r(n^{-7/4})X$.

Applying Lemma 2.2 to the graphs consisting of the added class-edges and deleted cross-edges, respectively, we get $X \leqslant (s^2 + s) + (h^2 + h)$. Therefore

$$\|E\boldsymbol{u}\|^2 \leqslant \frac{s^2 + h^2}{n} + O_r(n^{-1/2}). \tag{25}$$

Let $\boldsymbol{x}$ be an arbitrary unit vector. We write

$$\boldsymbol{x} := a\boldsymbol{u} + \boldsymbol{y}$$

with some vector $\boldsymbol{y}$ satisfying $\boldsymbol{y} \perp \boldsymbol{u}$, so that $a^2 + \beta^2 = 1$, where $\beta := \|\boldsymbol{y}\|$. Every eigenvalue of K other than μ is nonpositive, so $\boldsymbol{y}^T A_0 \boldsymbol{y} \leqslant 0$ and $\boldsymbol{x}^T A_0 \boldsymbol{x} \leqslant \mu a^2$. Moreover $\|E\| \leqslant \sqrt{2(s+h)} = O_r(n^{1/4})$. The norm bound gives $\boldsymbol{y}^T E \boldsymbol{y} \leqslant \|E\|\beta^2$, and the Cauchy–Schwarz inequality gives $|\boldsymbol{y}^T E \boldsymbol{u}| \leqslant \beta \|E\boldsymbol{u}\|$. Using $a^2 = 1 - \beta^2$ and $|a| \leqslant 1$, expanding $\boldsymbol{x}^T A(G) \boldsymbol{x}$ gives

$$\lambda(G) \leqslant \boldsymbol{x}^T A(G) \boldsymbol{x} \leqslant \mu + e_0 - (\mu + e_0 - \|E\|)\beta^2 + 2\beta\|E\boldsymbol{u}\|.$$

Combining with (22) and (24), we have $\mu + e_0 - \|E\| = \alpha n + O_r(n^{1/4}) > 0$. The maximizer $\beta = \|E\boldsymbol{u}\|/(\mu + e_0 - \|E\|) = O_r(n^{-3/4})$ lies in $[0,1]$. Maximizing over $\beta \geqslant 0$ gives

$$\lambda(G) \leqslant \mu + e_0 + \frac{\|E\boldsymbol{u}\|^2}{\mu + e_0 - \|E\|}. \tag{26}$$

Since $\mu + e_0 - \|E\| = \alpha n\big(1 + O_r(n^{-3/4})\big)$, using (25) gives

$$\frac{\|E\boldsymbol{u}\|^2}{\mu + e_0 - \|E\|} = \frac{(s^2+h^2)/n + O_r(n^{-1/2})}{\alpha n\big(1 + O_r(n^{-3/4})\big)} = \frac{(s^2+h^2)\alpha^{-1}}{n^2} + O_r(n^{-3/2}).$$

By Theorem 2.3 again, we have

$$\mu = \lambda(T_{n,r}) - \frac{2\alpha D}{n} + O(n^{-5/4}).$$

Substituting μ, e_0 and $\|E\boldsymbol{u}\|^2/(\mu + e_0 - \|E\|)$ into (26) gives the desired bound of $\lambda(G)$. □

3.2 Continuous supersaturation: Proof of Theorem 1.7

Next, we provide an immediate application of Theorem 1.8, showing that the spectral supersaturation in Theorem 1.7 is a direct consequence of Theorem 1.4 combined with Theorem 1.8.

Proof of Theorem 1.7. Let $\delta_F > 0$ be the constant in Theorem 1.4, and write $\alpha := 1 - \frac{1}{r}$. Adding q disjoint edges to $T_{n,r}$ raises the spectral radius by $2q/n$ up to lower-order terms. By Theorem 1.8, for every $q = O(\sqrt{n})$, we have

$$\lambda(Y_{n,r,q}) \leqslant \lambda(T_{n,r}) + \frac{2q}{n} + \frac{\alpha^{-1}q^2}{n^2} + O(n^{-5/4}). \tag{27}$$

Fix C with $0 < C \leqslant \delta_F/\sqrt{n}$, and set $q := \lfloor \frac{1}{2}Cn \rfloor - K$, where $K = K(r,F)$ is a constant integer fixed below. Then $q \leqslant \frac{1}{2}Cn \leqslant \frac{1}{2}\delta_F\sqrt{n}$. So $q = O(\sqrt{n})$ and (27) applies. Moreover, the two error terms in (27) are at most $\frac{1}{4n}\alpha^{-1}\delta_F^2 + O(n^{-5/4}) \leqslant K'/n$ for some absolute constant K'. Hence

$$\lambda(Y_{n,r,q}) \leqslant \lambda(T_{n,r}) + \frac{2}{n}\Big(\frac{Cn}{2} - K\Big) + \frac{K'}{n} = \lambda(T_{n,r}) + C - \frac{2K - K'}{n}.$$

Choosing $K \geqslant K'/2$ gives $\lambda(Y_{n,r,q}) \leqslant \lambda(T_{n,r}) + C \leqslant \lambda(G)$.

Set $\varepsilon_n := -(K+1)/n$. If $q \geqslant 1$, then $1 \leqslant q \leqslant \delta_F\sqrt{n}$, so applying Theorem 1.4 gives $\#F(G) \geqslant q\,c(n,F)$. Combining with $q \geqslant \frac{Cn}{2} - (K+1)$, we get $\#F(G) \geqslant \big(\frac{C}{2} + \varepsilon_n\big)n\,c(n,F)$. If $q < 1$, as K is an integer, then $\frac{Cn}{2} < K+1$, so $\big(\frac{C}{2} + \varepsilon_n\big)n = \frac{Cn}{2} - (K+1) < 0$ and the desired inequality holds trivially. In either case, we have $\#F(G) \geqslant \big(\frac{C}{2} - \frac{K+1}{n}\big)n\,c(n,F) = \big(\frac{C}{2} - o(1)\big)n\,c(n,F)$, as needed. □

4 The sharp threshold for cliques

In this section, we prove Theorem 1.5. To start with, we show the tightness of $\delta_r = (1 - \frac{1}{r})\sqrt{2}$.

Example 4.1. For any fixed $\varepsilon > 0$ and infinitely many n, let q be an integer with $(\delta_r + \varepsilon)\sqrt{n} \leqslant q = O(\sqrt{n})$, there exists an n-vertex graph H such that $\lambda(H) > \lambda(Y_{n,r,q})$ and $\#K_{r+1}(H) < q\,c(n,K_{r+1})$.

Proof. Let H be the graph defined in Proposition 2.5. Each copy of K_{r+1} using an added star edge must choose one vertex from each of the other $r-1$ parts. These other parts have sizes $m-1, m, \ldots, m$. Therefore, each added star edge lies in exactly $(m-1)m^{r-2}$ copies of K_{r+1}. Since a copy of K_{r+1} in an r-partite base graph plus internal edges has exactly one internal edge, these copies are counted once for each star edge. Hence

$$\#K_{r+1}(H) = q(m-1)m^{r-2} < qm^{r-1} = q\,c(n,K_{r+1}).$$

By Lemma 2.4, and since $T_{n,r}$ is exactly balanced when $n = rm$,

$$\lambda(Y_{n,r,q}) = \alpha n + \frac{2q}{n} + O_r(n^{-3/2}).$$

Applying Proposition 2.5, we obtain

$$\lambda(H) \geqslant \alpha n - \frac{2\alpha}{n} + \frac{2q}{n} + \frac{q^2\alpha^{-1}}{n^2} + O_r(n^{-3/2}).$$

Combining with the above two estimates, we get

$$\lambda(H) - \lambda(Y_{n,r,q}) \geqslant -\frac{2\alpha}{n} + \frac{q^2\alpha^{-1}}{n^2} + O_r(n^{-3/2}).$$

Since $q = (\delta_r + \varepsilon)\sqrt{n}$ and $\delta_r = \sqrt{2}(1 - \frac{1}{r}) = \sqrt{2}\alpha$, we have

$$\frac{q^2\alpha^{-1}}{n^2} = \alpha^{-1}(\delta_r + \varepsilon)^2\frac{1}{n} > \alpha^{-1}\delta_r^2\frac{1}{n} = \frac{2\alpha}{n}.$$

Thus, we obtain $\lambda(H) > \lambda(Y_{n,r,q})$ for all sufficiently large n. □

4.1 Stability and counting for cliques

Applying Theorem 2.1, we get the following structural reduction.

Lemma 4.2. *Fix $r \geqslant 2$, $\varepsilon > 0$, and let n be sufficiently large. Suppose that $1 \leqslant q \leqslant (\delta_r - \varepsilon)\sqrt{n}$ and suppose that G is an n-vertex graph satisfying*

$$\lambda(G) \geqslant \lambda(Y_{n,r,q})$$

and

$$\#K_{r+1}(G) \leqslant q\, c(n, K_{r+1}).$$

Then there exists $K = K_r(n_1, \dots, n_r)$ such that G is obtained from K by adding s class-edges and deleting h cross-edges, where $s, h = O_r(q)$. Moreover, $D := e(T_{n,r}) - e(K) = O_r(q)$.

Proof. By Lemma 2.4, we have $\lambda(Y_{n,r,q}) = \lambda(T_{n,r}) + 2q/n + O_r(n^{-3/2}) > \lambda(T_{n,r})$ for all sufficiently large n (using $q \geqslant 1$). Hence, the hypothesis gives $\lambda(G) \geqslant \lambda(T_{n,r})$. Since $q \leqslant (\delta_r - \varepsilon)\sqrt{n} \leqslant \delta_F n$ and $\#K_{r+1}(G) \leqslant q\, c(n, K_{r+1})$, applying Theorem 2.1 with $F = K_{r+1}$, we obtain a complete r-partite graph $K = K_r(n_1, \dots, n_r)$ from which G is built by adding s class-edges and deleting h cross-edges, where $h \leqslant 2s = O_r(q)$ and $|n_i - n_j| \leqslant 4\sqrt{s}$ for all $i \neq j$. Write $n_i = n/r + a_i$ with $\sum_i a_i = 0$. Since $|n_i - n_j| \leqslant 4\sqrt{s}$, each $|a_i| \leqslant 4\sqrt{s}$, so $B := \sum_i a_i^2 \leqslant 16rs = O_r(s)$. By (8), we get $D = \frac{1}{2}(B - B_T)$ with $B_T = O_r(1) \geqslant 0$, hence $D \leqslant \frac{1}{2}B = O_r(s) = O_r(q)$, as needed. □

The next lemma shows that, in the multipartite setting, each added class-edge creates about $c(n, K_{r+1})$ copies of K_{r+1}, up to losses from imbalance and from deleted cross-edges.

Lemma 4.3. *Let $K = K_r(n_1, \dots, n_r)$ be an n-vertex complete r-partite graph, and let G be obtained from K by adding s class-edges and deleting h cross-edges. Put $D := e(T_{n,r}) - e(K)$ and assume $s, h, D = O_r(\sqrt{n})$. Then there is a constant $C_r > 0$, depending only on r, such that*

$$\#K_{r+1}(G) \geqslant s\, c(n, K_{r+1}) - C_r\, s(\sqrt{D} + h + 1)\, n^{r-2}.$$

Proof. Let $V_1, \dots, V_r$ be the parts of K, $|V_i| = n_i$. Call a copy of K_{r+1} *canonical* if it has two vertices in one part and one vertex in each of the others. Its $r + 1$ vertices then form exactly one within-part pair, necessarily a class-edge, every other pair being a cross-edge. So each canonical copy uses exactly one class-edge, and it is a copy in G precisely when none of its cross-edges has been deleted. Hence, summing over the class-edges, we count each canonical copy once. We therefore count canonical copies that use only cross-edges of K and subtract those meeting a deleted cross-edge; as $\#K_{r+1}(G)$ is at least the number of survivors, the result follows.

Fix a class-edge $e \subseteq V_i$. Each canonical copy of K_{r+1} through e is determined by picking one vertex from each V_j with $j \neq i$, giving $\prod_{j\neq i} n_j$ of them. By (8), we have $\sum_i (n_i - \frac{n}{r})^2 = O_r(D)$, so $\max_i |n_i - \frac{n}{r}| = O_r(\sqrt{D})$. The sizes of parts of the Turán graph $T_{n,r}$ satisfy $t_i = \frac{n}{r} + O_r(1)$. Therefore, we conclude that

$$\prod_{j\neq i} n_j = \Big(\frac{n}{r}\Big)^{r-1} + O_r\big(\sqrt{D}\cdot n^{r-2}\big)$$

and

$$c(n, K_{r+1}) = \Big(\frac{n}{r}\Big)^{r-1} + O_r(n^{r-2}).$$

Therefore, we have $\prod_{j\neq i} n_j \geqslant c(n, K_{r+1}) - C_r(\sqrt{D}+1)n^{r-2}$. Summing over the s class-edges, the number of such canonical copies is at least $s\,c(n, K_{r+1}) - C_r\, s(\sqrt{D}+1)n^{r-2}$.

Fix a deleted cross-edge xy with $x \in V_a$, $y \in V_b$ ($a \neq b$), and a class-edge e; a canonical copy through e containing xy contains both x and y. If $e \subseteq V_a$, then x must be an endpoint of e and y the chosen vertex of V_b, leaving the $r-2$ parts other than V_a, V_b free (the case $e \subseteq V_b$ is symmetric); otherwise e, x, y are all determined, leaving $r-3$ parts free. Either way there are $O_r(n^{r-2})$ such copies. Summing over the s class-edges and h deleted cross-edges, at most $O_r(h\,s\,n^{r-2})$ canonical copies are lost. Consequently, we obtain $\#K_{r+1}(G) \geqslant s\,c(n, K_{r+1}) - C_r\, s(\sqrt{D}+1)n^{r-2} - O_r(h\,s\,n^{r-2})$, which after enlarging the constant C_r gives the desired bound of $\#K_{r+1}(G)$. □

4.2 Proof of Theorem 1.5

We prove the lower bound of Theorem 1.5. For any small $\varepsilon > 0$, suppose that $1 \leqslant q \leqslant (\delta_r - \varepsilon)\sqrt{n}$ and G is an n-vertex graph with $\lambda(G) \geqslant \lambda(Y_{n,r,q})$ and $\#K_{r+1}(G) \leqslant q\,c(n, K_{r+1})$. Our goal is to prove that $G \in \mathcal{T}_{n,r,q}$, that is, G is obtained from the Turán graph $T_{n,r}$ by adding q class-edges. By Lemma 4.2, G is obtained from a complete r-partite graph $K = K_r(n_1, \ldots, n_r)$ by adding s class-edges and deleting h cross-edges. Writing $D := e(T_{n,r}) - e(K) \geqslant 0$, we have

$$s, h, D = O_r(q) = O_r(\sqrt{n}).$$

Denote $\alpha := 1 - \frac{1}{r}$. Combining with Theorem 1.8, we get

$$\lambda(G) \leqslant \lambda(T_{n,r}) - \frac{2\alpha D}{n} + \frac{2(s-h)}{n} + \frac{(s^2+h^2)\alpha^{-1}}{n^2} + o(n^{-1}).$$

Lemma 2.4 gives $\lambda(Y_{n,r,q}) = \lambda(T_{n,r}) + 2q/n + O_r(n^{-3/2})$. Then

$$0 \leqslant \lambda(G) - \lambda(Y_{n,r,q}) \leqslant \frac{2(s-q-h)}{n} - \frac{2\alpha D}{n} + \frac{(s^2+h^2)\alpha^{-1}}{n^2} + o(n^{-1}). \tag{28}$$

Next, we divide the remaining arguments into three cases.

Case 1. $s \leqslant q-1$

Since $h = O_r(\sqrt{n})$, for large n, we get

$$-\frac{2h}{n} + \frac{h^2\alpha^{-1}}{n^2} = \frac{h}{n}\Big(\frac{h\alpha^{-1}}{n} - 2\Big) \leqslant 0.$$

Dropping the non-positive terms involving h and D, we see that (28) gives

$$\lambda(G) - \lambda(Y_{n,r,q}) \leqslant \frac{2(s-q)}{n} + \frac{s^2\alpha^{-1}}{n^2} + o(n^{-1}).$$

Since the right-hand side above is increasing in s and $q \leqslant (\delta_r - \varepsilon)\sqrt{n}$, we get

$$\lambda(G) - \lambda(Y_{n,r,q}) \leqslant -\frac{2}{n} + \frac{(q-1)^2\alpha^{-1}}{n^2} + o(n^{-1}) \leqslant \frac{-2 + (\delta_r - \varepsilon)^2\alpha^{-1}}{n} + o(n^{-1}) < 0,$$

where the last inequality holds since $\delta_r = \sqrt{2\alpha}$. This contradicts $\lambda(G) \geqslant \lambda(Y_{n,r,q})$.

Case 2. $s = q$

In this case, the inequality (28) reads

$$\lambda(G) - \lambda(Y_{n,r,q}) \leqslant \frac{-2h}{n} - \frac{2\alpha D}{n} + \frac{(q^2+h^2)\alpha^{-1}}{n^2} + o(n^{-1}).$$

If $h = 0$ and $D = 0$, then $e(K) = e(T_{n,r})$, so K is balanced, i.e., $K = T_{n,r}$. Thus, G is obtained from the Turán graph $T_{n,r}$ by adding q class-edges, which is the desired extremal case.

Next, we assume that $h \geqslant 1$ or $D \geqslant 1$. In either case, we have

$$-\frac{2h}{n} - \frac{2\alpha D}{n} + \frac{h^2\alpha^{-1}}{n^2} \leqslant -\frac{2\alpha}{n} + o(n^{-1}).$$

Indeed, if $D \geqslant 1$, this follows from $-2\alpha D/n \leqslant -2\alpha/n$ and $-2h/n + h^2\alpha^{-1}/n^2 \leqslant 0$. If $h \geqslant 1$, then $-2h/n + h^2\alpha^{-1}/n^2$ is decreasing on $h \in [1, O_r(\sqrt{n})]$, so its value is at most $-2/n + \alpha^{-1}/n^2$, which is at most $-2\alpha/n + o(n^{-1})$ because $\alpha = 1 - \frac{1}{r} < 1$. Therefore, we get

$$\lambda(G) - \lambda(Y_{n,r,q}) \leqslant -\frac{2\alpha}{n} + \frac{q^2\alpha^{-1}}{n^2} + o(n^{-1}) < 0$$

for large n, again a contradiction.

Case 3. $s = q + \ell$ **with** $\ell \geqslant 1$

The assumption gives $\#K_{r+1}(G) \leqslant q\, c(n, K_{r+1})$. By Lemma 4.3, we have

$$\#K_{r+1}(G) \geqslant s\, c(n, K_{r+1}) - C_r\, s(\sqrt{D} + h + 1)n^{r-2}.$$

Combining with the two bounds, we obtain $\ell\, c(n, K_{r+1}) \leqslant C_r\, s(\sqrt{D} + h + 1)n^{r-2}$, which together with $c(n, K_{r+1}) = (n/r)^{r-1} + O_r(n^{r-2})$ yields $\sqrt{D} + h + 1 \geqslant \kappa\frac{\ell n}{s}$ for some $\kappa = \kappa(r) > 0$. Since $D = e(T_{n,r}) - e(K)$ is a non-negative integer, we have $D \geqslant \sqrt{D}$. Then

$$\frac{2h + 2\alpha D}{n} \geqslant \frac{2\alpha h + 2\alpha\sqrt{D}}{n} \geqslant \frac{2\alpha\kappa\ell}{s} - \frac{2\alpha}{n}.$$

Substituting into (28), where $s - q = \ell$ and $\frac{(s^2+h^2)\alpha^{-1}}{n^2} = O_r(n^{-1})$, we get

$$\lambda(G) - \lambda(Y_{n,r,q}) \leqslant \frac{2\ell}{n} - \frac{2\alpha\kappa\ell}{s} + O_r(n^{-1}).$$

As $s = O_r(\sqrt{n})$ and $\ell \geqslant 1$, we have $\frac{2\ell}{n} < \frac{\alpha\kappa\ell}{s}$ for large n, so $\frac{2\ell}{n} - \frac{2\alpha\kappa\ell}{s} < -\frac{\alpha\kappa\ell}{s}$. This term is of order $n^{-1/2}$ and dominates the $O_r(n^{-1})$ remainder, so $\lambda(G) < \lambda(Y_{n,r,q})$ for large n, a contradiction.

In conclusion, we obtain that for every integer $1 \leqslant q \leqslant (\delta_r - \varepsilon)\sqrt{n}$, if G is a graph with $\lambda(G) \geqslant \lambda(Y_{n,r,q})$ and $\#K_{r+1}(G) \leqslant q\, c(n, K_{r+1})$, then $s = q$, and G is obtained from $T_{n,r}$ by adding q edges. Each class-edge added to $T_{n,r}$ lies in at least $c(n, K_{r+1})$ copies of K_{r+1}, and distinct class-edges give distinct copies. Hence, we conclude that $G \in \mathcal{T}_{n,r,q}$ and $\#K_{r+1}(G) = q\, c(n, K_{r+1})$.

5 The sharp threshold for odd cycles

First of all, we show that the threshold $\delta_{\mathrm{opt}} = 1/\sqrt{2}$ in Theorem 1.6 is best possible.

Example 5.1. Fix $\varepsilon > 0$ and let N be sufficiently large. Let q be an integer with $(1+\varepsilon)\sqrt{N} \leqslant q \leqslant C\sqrt{N}$ for some constant C, and let $H_{N,q}$ be obtained from $K_{N+1,N-1}$ by adding a star with q edges inside the part of size $N+1$. Then $\lambda(H_{N,q}) > \lambda(Y_{2N,2,q})$ and $\#C_{2k+1}(H_{N,q}) < q\, c(2N, C_{2k+1})$.

Proof. By Proposition 2.5, we have

$$\lambda(H_{N,q}) \geqslant N + \frac{q}{N} + \frac{q^2 - N}{2N^2} + O(N^{-3/2}).$$

Applying Lemma 2.4, it follows that

$$\lambda(Y_{2N,2,q}) = N + \frac{q}{N} + O(N^{-3/2}).$$

Therefore, we get

$$\lambda(H_{N,q}) - \lambda(Y_{2N,2,q}) \geqslant \frac{q^2 - N}{2N^2} + O(N^{-3/2}).$$

If $q \geqslant (1+\varepsilon)\sqrt{N}$, then $q^2 - N \geqslant \big(2\varepsilon + \varepsilon^2\big)N$. So the main term above is a positive constant times N^{-1}, while the error term is $O(N^{-3/2})$. Thus, $\lambda(H_{N,q}) > \lambda(Y_{2N,2,q})$ for all sufficiently large N.

Write L and R for the two parts of $K_{N+1,N-1}$, with $|L| = N+1$ and $|R| = N-1$. The q star edges lie in L and share a common center v. Call an edge of $H_{N,q}$ a *cross edge* if it joins L and R, and a *star edge* otherwise. First, we see that each copy of C_{2k+1} uses exactly one star edge. Walk once around such a cycle. A cross edge switches the side between L and R, while a star edge stays on the same side. Since the walk returns to its start, the number of switches is even, so the cycle uses an even number of cross edges. As the length equals the number of cross edges plus the number of star edges, the cycle uses an *odd* number of star edges. On the other hand, every vertex of a cycle has degree 2, so the center v lies on at most two of the used star edges; hence the cycle uses at most two. The odd number that is at most two equals one.

Fix a star edge xy, with $x, y \in L$. A copy of C_{2k+1} through xy consists of xy together with a path of length $2k$ from x to y alternating between L and R: $x, b_1, a_1, b_2, \ldots, a_{k-1}, b_k, y$, where $b_1, \ldots, b_k \in R$ are distinct and $a_1, \ldots, a_{k-1} \in L \setminus \{x, y\}$ are distinct. Each such copy corresponds to exactly one such sequence (read from x to y). A sequence is fixed by an ordered choice of the b_i from R ($N-1$ vertices) and an ordered choice of the a_i from $L \setminus \{x, y\}$ (also $N-1$ vertices), giving $(N-1)_k\,(N-1)_{k-1}$ copies through xy. Summing over the q star edges, we get

$$\#\,C_{2k+1}(H_{N,q}) = q\,(N-1)_k\,(N-1)_{k-1}.$$

Similarly, by adding an edge inside a part of $K_{N,N}$, we have

$$c(2N, C_{2k+1}) = (N)_k\,(N-2)_{k-1}.$$

Hence

$$\frac{(N-1)_k\,(N-1)_{k-1}}{(N)_k\,(N-2)_{k-1}} = \frac{N-k}{N} \cdot \frac{N-1}{N-k} = \frac{N-1}{N} < 1,$$

so $(N-1)_k(N-1)_{k-1} < c(2N, C_{2k+1})$. Multiplying by q gives $\#\,C_{2k+1}(H_{N,q}) < q\,c(2N, C_{2k+1})$. □

5.1 Stability and counting for odd cycles

Applying Theorem 2.1, we get the following stability reduction.

Lemma 5.2. *Fix $\varepsilon > 0$ and suppose that $n = 2N + \sigma$, where $\sigma \in \{0, 1\}$, and q is an integer such that $1 \leqslant q \leqslant (1-\varepsilon)\sqrt{N}$. If G is an n-vertex graph with*

$$\lambda(G) \geqslant \lambda(Y_{2N+\sigma,2,q}),$$

and

$$\#C_{2k+1}(G) \leqslant q\,c(n, C_{2k+1}),$$

then there is a partition $V(G) = A \sqcup B$ such that G is obtained from the complete bipartite graph $K_{|A|,|B|}$ by adding s edges inside the two parts and deleting h cross-edges, where $s, h = O_k(q)$. Moreover, if we write $|A| = N + \sigma + a$ and $|B| = N - a$, where $a \geqslant 0$, then $a = O_k(\sqrt{s})$.

Proof. Let δ_F, η_F be the constants of Theorem 2.1 for $F = C_{2k+1}$ and, say, $\varepsilon_1 = \frac{1}{4}$. Clearly, we have $\lambda(G) \geqslant \lambda(Y_{n,2,q}) \geqslant \lambda(T_{n,2})$. Moreover, $q \leqslant (1-\varepsilon)\sqrt{N} \leqslant \delta_F n$ and $\#\,C_{2k+1}(G) \leqslant q\,c(n, C_{2k+1}) \leqslant (q+\varepsilon_1)\,c(n, C_{2k+1})$. Applying Theorem 2.1 with $r = 2$, we get a partition $V(G) = A \sqcup B$ such that G

is obtained from $K_{|A|,|B|}$ by adding s class-edges and deleting h cross-edges, where $h \leqslant 2s \leqslant 4q/\eta_F$, so $s, h = O_k(q)$, and $\big||A|-|B|\big| \leqslant 4\sqrt{s}$. We may assume $|A| \geqslant |B|$, so $|A| \geqslant \lceil n/2 \rceil = N+\sigma$. Writing $|A| = N+\sigma+a$ and $|B| = N-a$, we get $a \geqslant 0$ and $2a+\sigma = |A|-|B| \leqslant 4\sqrt{s}$, hence $a \leqslant 2\sqrt{s}$. □

The following result is a direct consequence of Theorem 1.8 in the case $r = 2$.

Lemma 5.3. *Let $n = 2N+\sigma$, where $\sigma \in \{0,1\}$. Let G be obtained from $K_{p,b}$, where $p = N+\sigma+a$ and $b = N-a$, by adding s edges inside the two parts and deleting h cross-edges. Assume*

$$a = O_k(N^{1/4}), \qquad s = O_k(\sqrt{N}), \qquad h = O_k(\sqrt{N}).$$

Then

$$\lambda(G) \leqslant \sqrt{N(N+\sigma)} + \frac{s-h}{N} - \frac{a(a+\sigma)}{2N} + \frac{s^2+h^2}{2N^2} + o(N^{-1}).$$

Proof. Apply Theorem 1.8 with $r = 2$ and $\alpha = \frac{1}{2}$ to $K = K_{p,b}$, where $n = p+b = 2N+\sigma$. Note that $\lambda(T_{n,2}) = \sqrt{N(N+\sigma)}$ exactly, since $T_{n,2} = K_{N+\sigma,N}$. Also

$$D = e(T_{n,2}) - e(K_{p,b}) = N(N+\sigma) - (N+\sigma+a)(N-a) = a(a+\sigma),$$

so $D = O_k(\sqrt{N})$ by $a = O_k(N^{1/4})$, and the hypotheses of Theorem 1.8 hold. The theorem yields

$$\lambda(G) \leqslant \sqrt{N(N+\sigma)} - \frac{a(a+\sigma)}{2N+\sigma} + \frac{2(s-h)}{2N+\sigma} + \frac{2(s^2+h^2)}{(2N+\sigma)^2} + O_k(N^{-5/4}).$$

Since $a(a+\sigma)$, $s+h = O_k(\sqrt{N})$ and $\sigma \in \{0,1\}$, replacing $2N+\sigma$ by $2N$ in each term changes it by at most $O_k(N^{-3/2})$, which is absorbed into $o(N^{-1})$. This gives the stated bound. □

The next lemma is the odd-cycle analogue of Lemma 4.3.

Lemma 5.4. *Under the condition of Lemma 5.3, there exists a constant $C'_k > 0$ such that*

$$\#C_{2k+1}(G) \geqslant s\, c(2N+\sigma, C_{2k+1}) - C'_k s(a+h+1)N^{2k-2}.$$

Proof. We follow the proof of Lemma 4.3, replacing the product over the other parts by the alternating-path count of Example 5.1. Call a copy of C_{2k+1} *canonical* if it uses exactly one class-edge. By the parity argument of Example 5.1, every copy uses at least one class-edge, so it suffices to lower bound the surviving canonical copies, and each is counted once by its unique class-edge.

Fix a class-edge $e = xy$ in a part W of $K_{p,b}$, with opposite part Z. The canonical copies through e are e plus an alternating x–y path of length $2k$, of which there are $(|Z|)_k(|W|-2)_{k-1}$ in $K_{p,b}$. Since $a = O_k(N^{1/4})$, every part has size $N+O_k(N^{1/4})$, so this equals $c(2N+\sigma, C_{2k+1}) + O_k((a+1)N^{2k-2})$.

A deleted cross-edge uv kills only canonical copies that use it as a path-edge; fixing its position and an endpoint leaves $\leqslant 2k-2$ free vertices, so at most $O_k(N^{2k-2})$ copies through each e. Summing over the s class-edges and h deleted cross-edges and enlarging the constant gives

$$\#C_{2k+1}(G) \geqslant s\, c(2N+\sigma, C_{2k+1}) - C'_k\, s(a+h+1)N^{2k-2}.$$ □

5.2 Proof of Theorem 1.6

Since $\sqrt{n} = \sqrt{2N+\sigma} = \sqrt{2N}(1+o(1))$, the condition $q \leqslant (1/\sqrt{2}-\varepsilon)\sqrt{n}$ implies $q \leqslant (1-\varepsilon')\sqrt{N}$ for some $\varepsilon' > 0$, so it suffices to show that for any fixed $k \geqslant 1$ and $\varepsilon > 0$, and for all sufficiently large N, if $q \leqslant (1-\varepsilon)\sqrt{N}$ and G is a graph of order $n = 2N+\sigma$ (where $\sigma \in \{0,1\}$) with $\lambda(G) \geqslant \lambda(Y_{n,2,q})$ and $\#C_{2k+1}(G) \leqslant q\, c(n, C_{2k+1})$, then G is obtained from $T_{n,2}$ by adding q edges. First, Lemma 5.2 implies that G admits a partition $V(G) = A \sqcup B$ satisfying $|A| = N+\sigma+a$ and $|B| = N-a$, where $a = O_k(N^{1/4})$. Moreover, writing $s = e(A)+e(B)$ and $h = |A||B| - e(A,B)$, we have $s, h = O(\sqrt{N})$. Combining Lemma 5.3 and Lemma 2.4 gives

$$\lambda(G) - \lambda(Y_{n,2,q}) \leqslant \frac{s-q-h}{N} - \frac{a(a+\sigma)}{2N} + \frac{s^2+h^2}{2N^2} + o(N^{-1}). \tag{29}$$

We now split the remaining argument into three cases.

Case 1. $s \leqslant q-1$

Since $h = O(\sqrt{N})$, for large N, we have $-\frac{h}{N} + \frac{h^2}{2N^2} \leqslant 0$. Also $a(a+\sigma) \geqslant 0$. Thus the right side of (29) is maximized by taking $s = q-1$, $h = 0$, and $a = 0$. Therefore

$$\lambda(G) - \lambda(Y_{n,2,q}) \leqslant -\frac{1}{N} + \frac{(q-1)^2}{2N^2} + o(N^{-1}) \leqslant -\frac{1}{N} + \frac{q^2}{2N^2} + o(N^{-1}).$$

By the assumption $q \leqslant (1-\varepsilon)\sqrt{N}$, we have $\frac{q^2}{2N^2} \leqslant \frac{(1-\varepsilon)^2}{2N}$. Hence

$$\lambda(G) - \lambda(Y_{n,2,q}) \leqslant \left(-1 + \frac{(1-\varepsilon)^2}{2}\right)\frac{1}{N} + o(N^{-1}) < 0,$$

which contradicts the assumption $\lambda(G) \geqslant \lambda(Y_{n,2,q})$.

Case 2. $s = q$

If $a = 0$ and $h = 0$, then G is obtained from $K_{N+\sigma,N}$ by adding $s = q$ edges. This is the desired extremal case. Next, we consider the case $a + h \geqslant 1$. Since $s = q$, we get from (29) that

$$\lambda(G) - \lambda(Y_{n,2,q}) \leqslant -\frac{h}{N} - \frac{a(a+\sigma)}{2N} + \frac{q^2+h^2}{2N^2} + o(N^{-1}).$$

Recall that $h = O(\sqrt{N})$. We have $-\frac{h}{N} + \frac{h^2}{2N^2} \leqslant -\frac{h}{2N}$ for large N. Since a and h are non-negative integers and $a + h \geqslant 1$, we get $\frac{h}{2} + \frac{a(a+\sigma)}{2} \geqslant \frac{1}{2}$. Therefore, it follows that

$$\lambda(G) - \lambda(Y_{n,2,q}) \leqslant -\frac{1}{2N} + \frac{q^2}{2N^2} + o(N^{-1}).$$

Since $q \leqslant (1-\varepsilon)\sqrt{N}$, we get $-\frac{1}{2N} + \frac{q^2}{2N^2} \leqslant -\frac{1-(1-\varepsilon)^2}{2N} < 0$. This is again a contradiction.

Case 3. $s \geqslant q+1$

Write $s = q + \ell$ with $\ell \geqslant 1$. The assumption gives $\#\,C_{2k+1}(G) \leqslant qc(n, C_{2k+1})$, while Lemma 5.4 gives $\#\,C_{2k+1}(G) \geqslant sc(n, C_{2k+1}) - C'_k s(a+h+1)N^{2k-2}$. Combining the two, we obtain

$$qc(n, C_{2k+1}) \geqslant (q+\ell)c(n, C_{2k+1}) - C'_k s(a+h+1)N^{2k-2},$$

so $\ell c(n, C_{2k+1}) \leqslant C'_k s(a+h+1)N^{2k-2}$. Note that $c(n, C_{2k+1}) = \Omega_k(N^{2k-1})$ for sufficiently large N. Substituting this and dividing by $C'_k s N^{2k-2}$ yields a constant $c_k > 0$ satisfying

$$a + h \geqslant \frac{c_k \ell N}{s} - 1.$$

We now bound the negative terms in the spectral estimate (29). Since $a(a+\sigma) \geqslant a$ when $a \geqslant 0$, we have $h + \frac{a(a+\sigma)}{2} \geqslant \frac{a+h}{2} \geqslant \frac{1}{2}\big(c_k \frac{\ell N}{s} - 1\big)$. Dividing by N gives

$$\frac{h}{N} + \frac{a(a+\sigma)}{2N} \geqslant \frac{c_k}{2}\frac{\ell}{s} - \frac{1}{2N}.$$

Applying (29) with $s - q = \ell$, we have

$$\lambda(G) - \lambda(Y_{n,2,q}) \leqslant \frac{\ell}{N} - \frac{c_k}{2}\frac{\ell}{s} + \frac{1}{2N} + \frac{s^2+h^2}{2N^2} + o(N^{-1}).$$

By the structural reduction $s, h = O_k(\sqrt{N})$, so the last three terms are $O_k(N^{-1})$ and $s \leqslant C''_k\sqrt{N}$ for some constant $C''_k > 0$. For large N, we have $\frac{c_k}{2s} - \frac{1}{N} \geqslant \frac{c_k}{4s}$. Hence

$$\frac{\ell}{N} - \frac{c_k}{2}\frac{\ell}{s} = -\ell\Big(\frac{c_k}{2s} - \frac{1}{N}\Big) \leqslant -\frac{\ell c_k}{4s} \leqslant -\frac{c_k}{4C''_k\sqrt{N}}.$$

Therefore, we obtain

$$\lambda(G) - \lambda(Y_{n,2,q}) \leqslant -\Omega_k(N^{-1/2}) + O_k(N^{-1}) < 0$$

for large N, contradicting $\lambda(G) \geqslant \lambda(Y_{n,2,q})$. Thus, Case 3 is impossible.

In conclusion, if $\#C_{2k+1}(G) \leqslant q\, c(n, C_{2k+1})$, then $s = q$ and $a = h = 0$, so G is obtained from $T_{n,2}$ by adding q class-edges. Each class-edge lies in at least $c(n, C_{2k+1})$ copies of C_{2k+1}, and distinct class-edges give distinct copies. Hence, we get $G \in \mathcal{T}_{n,2,q}$ and $\#C_{2k+1}(G) \geqslant q\, c(n, C_{2k+1})$.

6 Concluding remarks

In this paper, we studied the spectral supersaturation problems for cliques and odd cycles, and determined the exact thresholds $\delta_{\mathrm{spec}}(K_{r+1})$ and $\delta_{\mathrm{spec}}(C_{2k+1})$. We proved that the clique threshold is $(1-\frac{1}{r})\sqrt{2}$ for every $r \geqslant 2$ (Theorem 1.5), and the odd-cycle threshold is the universal value $1/\sqrt{2}$ for every $k \geqslant 1$ (Theorem 1.6). These are the spectral supersaturation results with both the exact count $q\, c(n, F)$ and the optimal range of q. A common theme of these spectral results is the sharp contrast with the edge setting of Lovász–Simonovits and Pikhurko–Yilma in Theorems 1.2 and 1.3, respectively. For example, the range of q drops down from order n to order $\sqrt{n}$, and the clique threshold changes from $1/r$ to $(1-\frac{1}{r})\sqrt{2}$. The proofs rely on the spectral supersaturation-stability method, together with a second-order spectral estimate (Theorem 1.8) for nearly Turán graphs, which controls the spectral radius of a complete r-partite graph after $O_r(\sqrt{n})$ edge changes and may be of independent interest. As a further application, we proved a spectral supersaturation with a continuous additive gap (Theorem 1.7) that bounds $\#F(G)$ by the spectral increment $\lambda(G)-\lambda(T_{n,r})$. The clique case improves the Bollobás–Nikiforov bound just above the Turán threshold.

Building on these results, we conclude with some open problems for further study.

Determining thresholds for all color-critical graphs. Recall that $\delta_{\mathrm{spec}}(F)$ is defined to be the largest constant δ_F such that for every $\varepsilon > 0$, all sufficiently large n and every integer $1 \leqslant q \leqslant (\delta_F - \varepsilon)\sqrt{n}$, each n-vertex graph G with $\lambda(G) \geqslant \lambda(Y_{n,r,q})$ contains at least $q\, c(n, F)$ copies of F. We conjecture that the spectral threshold $\delta_{\mathrm{spec}}(F)$ depends on F only through its chromatic number, and this threshold may be governed by whether a single added structure can raise the spectral radius above $\lambda(Y_{n,r,q})$. This would contrast with the edge setting, where the threshold $\delta_{\mathrm{edge}}(F)$ does depend on the structural properties of F: it equals $1/r$ for K_{r+1}, but $(r-1)/r^2$ for $K_{r+2} - e$, the clique with one edge removed [36]. A spectral threshold independent of the structure of F would be a different phenomenon. The graph $K_{r+2} - e$ is the natural first test case. We speculate that $K_{r+2} - e$ may have the same spectral threshold as K_{r+1} despite a different edge threshold.

The linear range of the increment. Our results for cliques and odd cycles are located in the range $q < \delta_F\sqrt{n}$. It would be interesting to describe the behavior above the threshold, say $q \gg \sqrt{n}$, where the candidate graph $Y_{n,r,q}$ is no longer extremal, and the intermediate range $\sqrt{n} \ll q \ll n$ is widely open. Since the condition $\lambda(G) \geqslant \lambda(Y_{n,r,q})$ no longer forces $\#F(G) \geqslant q\, c(n, F)$ in this range, it would be worth determining the new candidate graph and the correct order of $\#F(G)$.

Let $T_{n,r,q}$ be obtained from Turán's graph $T_{n,r}$ by adding a star with q edges into a largest part. Fang, Li, Lin and Ma [12, Theorem 7.2] showed that the range of q can be extended to be linear when taking $T_{n,r,q}$ as a candidate graph, that is, there is a constant $\delta_F^* > 0$ such that for every integer $1 \leqslant q \leqslant \delta_F^* n$, the condition $\lambda(G) \geqslant \lambda(T_{n,r,q})$ forces $\#F(G) \geqslant q\, c(n, F)$. It is interesting to determine the new threshold δ_F^* for every color-critical graph F. We conjecture that the threshold δ_F^* for cliques and odd cycles would be consistent with those in the edge setting; see Table 1.

For Theorem 1.7, we showed a continuous version for counting color-critical graphs in any graph G with $\lambda(G) \geqslant \lambda(T_{n,r}) + C$, where $0 < C \leqslant \delta_F n^{-1/2}$. Naturally, it is interesting to count substructures in the host graph with arbitrary spectral density above $\lambda(T_{n,r})$. For example, one could study the spectral supersaturation in the remaining range $C \gg n^{-1/2}$. The problem of counting cliques under the edge density condition was already solved by Reiher [38].

Counting degenerate bipartite graphs. Our framework is tailored to color-critical graphs, for which the Turán number is $\Theta(n^2)$ and $T_{n,r}$ is the unique extremal graph; this is what makes $Y_{n,r,q}$ the right comparison graph. For degenerate F, such as C_{2t} or $K_{t,t}$, the Turán number is $o(n^2)$ and the extremal graphs are sparse, so neither $T_{n,r}$ nor $Y_{n,r,q}$ has an obvious analogue, and even the correct order of the threshold q is unclear. Building a spectral supersaturation theory for degenerate F, relative to the appropriate sparse extremal structures, appears to require new ideas.

Here, we present a possible problem for counting C_{2t}. Let

$$S^+_{t-1,n} := K_{t-1} \vee I^+_{n-t+1}$$

be the n-vertex graph obtained from the join $K_{t-1} \vee I_{n-t+1}$ by adding an edge into its independent set. A breakthrough of Cioabă, Desai and Tait [7] shows that for any fixed $t \geqslant 3$, if n is sufficiently large and G is an n-vertex graph with $\lambda(G) > \lambda(S^+_{t-1,n})$, then G contains a $2t$-cycle. We refer the readers to [4] for a general spectral theorem for degenerate bipartite graphs. Inspired by the above result, we propose the following spectral supersaturation problem for C_{2t} for every $t \geqslant 3$.

Problem 6.1 (Counting even cycles)**.** *For every integer $t \geqslant 3$, does there exist a constant $c(t) > 0$ such that if n is sufficiently large and G is an n-vertex graph with $\lambda(G) > \lambda(S^+_{t-1,n})$, then*

$$\#C_{2t}(G) \;\geqslant\; c(t) \cdot n^{t-3} \;?$$

If the above is true, what is the largest value of such $c(t)$?

Finally, a complementary direction replaces the vertex-spectral condition $\lambda(G) \geqslant \lambda(T_{n,r})$ by the edge-spectral condition $\lambda^2(G) \geqslant (1 - \frac{1}{r})2m$, in the spirit of Nikiforov's theorem [28]. For counting triangles and books, this scale is the natural one and has been well-studied; we refer the readers to [34, 44, 45, 46]. It would be interesting to develop the corresponding supersaturation counts and to connect both the two scales for general graphs F; see [17, 18, 19, 21] for recent developments.

Acknowledgment

The corresponding author would like to thank Prof. Bo Ning for raising the optimal spectral supersaturation problem after a talk at Anhui University in May 2025. Thanks also go to Dr. Longfei Fang for helpful discussions during the study of the spectral graph problem over the past few years. In an early exploratory stage of the project, the authors additionally used language-model-based tools to brainstorm candidate proof strategies. The authors developed the arguments, wrote and verified all proofs. Any suggestions arising from these tools served only as informal inspiration.

References

[1] J. Balogh, F.C. Clemen, On stability of the Erdős–Rademacher problem, Illinois J. Math. 67 (1) (2023) 1–11.

[2] B. Bollobás, Extremal Graph Theory, Academic Press, New York, 1978.

[3] B. Bollobás, V. Nikiforov, Cliques and the spectral radius, J. Combin. Theory Ser. B 97 (2007) 859–865.

[4] J. Byrne, D.N. Desai, M. Tait, A general theorem in spectral extremal graph theory, Trans. Amer. Math. Soc. (2026), see https://doi.org/10.1090/tran/9691

[5] H. Chen, Y. Li, An edge-spectral supersaturation of Mubayi's theorem for color-critical graphs, (2026), arXiv:2607.01073.

[6] H. Chen, Y. Li, Q. Tang, Supersaturation in Nosal graphs: Triangles and books, (2026), arXiv:2607.16746.

[7] S. Cioabă, D. Desai, M. Tait. The spectral even cycle problem. Combinatorial Theory, 4 (1) (2024), Paper No.10.

[8] P. Erdős, On a theorem of Rademacher–Turán, Illinois J. Math. 6 (1962) 122–127.

[9] P. Erdős, On the number of triangles contained in certain graphs, Canad. Math. Bull. 7 (1) (1964) 53–56.

[10] P. Erdős, M. Simonovits, Supersaturated graphs and hypergraphs, Combinatorica 3 (2) (1983) 181–192.

[11] L. Kang, V. Nikiforov, Extremal problems for the p-spectral radius of graphs, Electron. J. Combin. 21 (2014), Paper No. 3.21.

[12] L. Fang, Y. Li, H. Lin, J. Ma, Spectral supersaturation for color-critical graphs, (2025), arXiv:2512.22482.

[13] L. Fang, H. Lin, M. Zhai, Counting color-critical subgraphs under Nikiforov's condition, (2026), arXiv:2603.14964.

[14] L. Fang, H. Lin, M. Zhai, Edge-spectral supersaturation for tripartite color-critical graphs, (2026), arXiv:2608.04485.

[15] Y. Li, L. Feng, Y. Peng, A spectral Lovász–Simonovits theorem, (2024), arXiv:2408.01709.

[16] Y. Li, H. Liu, S. Zhang, More on Nosal's spectral theorem: Books and 4-cycles, J. Combin. Theory Ser. B 179 (2026) 219–249.

[17] Y. Li, H. Liu, S. Zhang, An edge-spectral Erdős–Stone–Simonovits theorem and its stability, (2025), arXiv:2508.15271.

[18] Y. Li, H. Liu, S. Zhang, Edge-spectral Turán theorems for color-critical graphs with applications, (2025), SIAM J. Discrete Math., to appear, arXiv:2511.15431.

[19] Y. Li, W. Lin, H. Liu, S. Zhang, Spectral Sidorenko inequalities and edge-spectral supersaturation, (2026), arXiv:2605.26614.

[20] H. Liu, O. Pikhurko, K. Staden, The exact minimum number of triangles in graphs of given order and size, Forum of Math. Pi 8, No. e8, (2020), 144 pages.

[21] J. Liu, Z. Lou, S. Sun, Sharp edge-spectral supersaturation for odd cycles, (2026), arXiv:2609.08077.

[22] L. Liu, B. Ning, On spectral Turán theorems: Confirming a conjecture of Guiduli and two problems of Nikiforov, (2026), arXiv:2605.05048.

[23] X. Liu, D. Mubayi, On a generalized Erdős–Rademacher problem, J. Graph Theory 100 (2022) 101–126.

[24] L. Lovász, M. Simonovits, On the number of complete subgraphs of a graph, in: Proc. of Fifth British Comb. Conf., Aberdeen, 1975, pp. 431–442.

[25] L. Lovász, M. Simonovits, On the number of complete subgraphs of a graph II, in: Studies in Pure Math, Birkhäuser (dedicated to P. Turán), 1983, pp. 459–495.

[26] J. Ma, L.-T. Yuan, Supersaturation beyond color-critical graphs, Combinatorica 45 (2) (2025), Paper No. 18.

[27] D. Mubayi, Counting substructures I: Color critical graphs, Adv. Math. 225 (2010) 2731–2740.

[28] V. Nikiforov, Some inequalities for the largest eigenvalue of a graph, Combin. Probab. Comput. 11 (2002), 179–189.

[29] V. Nikiforov, A spectral Erdős–Stone–Bollobás theorem, Combin. Probab. Comput. 18 (2009) 455–458.

[30] V. Nikiforov, Stability for large forbidden subgraphs, J. Graph Theory 62 (4) (2009) 362–368.

[31] V. Nikiforov, Spectral saturation: Inverting the spectral Turán theorem, Electron. J. Combin., 16 (1) (2009), Paper No. 33.

[32] V. Nikiforov, The number of cliques in graphs of given order and size, Trans. Amer. Math. Soc. 363 (3) (2011) 1599–1618.

[33] V. Nikiforov, Some new results in extremal graph theory, in: Surveys in Combinatorics 2011, London Math. Soc. Lecture Note Ser. 392, Cambridge Univ. Press, 2011, pp. 141–181.

[34] B. Ning, M. Zhai, Counting substructures and eigenvalues I: Triangles, European J. Combin. 110 (2023), No. 103685.

[35] B. Ning, M. Zhai, Counting substructures and eigenvalues II: Quadrilaterals, Electron. J. Combin. 32 (4) (2025), #P4.1

[36] O. Pikhurko, Z. Yilma, Supersaturation problem for color-critical graphs, J. Combin. Theory Ser. B 123 (2017) 148–185.

[37] A.A. Razborov, On the minimal density of triangles in graphs, Combin. Probab. Comput. 17 (4) (2008) 603–618.

[38] C. Reiher, The clique density theorem, Ann. of Math. 184 (3) (2016) 683–707.

[39] M. Simonovits, A method for solving extremal problems in graph theory, stability problems, in: Theory of Graphs, Proc. Colloq., Tihany, 1966, Academic Press, New York, (1968), pp. 279–319.

[40] B. Wang, Z. Lou, J. Shu, On a spectral booksize problem for non-bipartite graphs, (2026), arXiv:2608.05947.

[41] H. Wilf, Spectral bounds for the clique and independence numbers of graphs, J. Combin. Theory Ser. B 40 (1986) 113–117.

[42] C. Xiao, G.O. Katona, The number of triangles is more when they have no common vertex, Discrete Math. 344 (2021), No. 112330.

[43] M. Zhai, H. Lin, A strengthening of the spectral color critical edge theorem: Books and theta graphs, J. Graph Theory 102 (3) (2023) 502–520.

[44] M. Zhai, R. Li, Z. Lou, Advances on two spectral conjectures regarding booksize of graphs, European J. Combin. 138 (2026), No.104431.

[45] Y. Zhang, M. Zhai, A spectral threshold for triangle counting, (2026), arXiv:2606.08163.

[46] X. Zhao, L. You, J. Zeng, X. Zhang, Two problems on booksize and triangular edges in Nosal graphs, (2026), arXiv:2607.15071.